Support Vector Machines

[SVM] 1-6

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Perceptron Learning Algorithm (PLA)

- Binary Classification:
  - Features: $x=(1,x_1,\ldots,x_d)$
  - Labels: $\{-1,1\}$

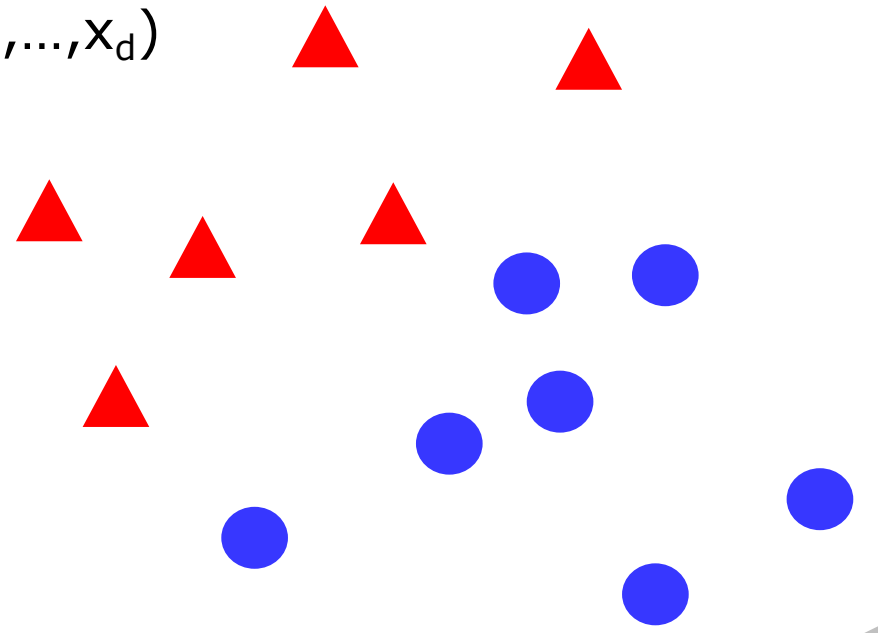
- Linear Model:
  - Parameters of hypothesis w: $(w_0, w_1,\ldots, w_d)$
  - Predict: $w(x) = \text{sign}(w_0 1 + w_1 x_1 + \ldots + w_d x_d) = \text{sign}(w^T x)$

- Perceptron:
  - Algorithm for finding a *good* set of parameters $w_0,\ldots,w_d$.
  - Good: Classifies (most of) training data correctly.
  - Recall: Hope good performance on training data means good performance on new data.



Perceptron Learning Algorithm (PLA)

- Surprisingly simple algorithm!
  - Features: $x=(x_0,x_1,\ldots,x_d)$
  - Labels: {-1,1}



- Algorithm:
  - w = (0,0,…,0)
  - While there is data point (x,y) with $y \neq \text{sign}(w^T x)$:
    - w = w + yx
  - Return w

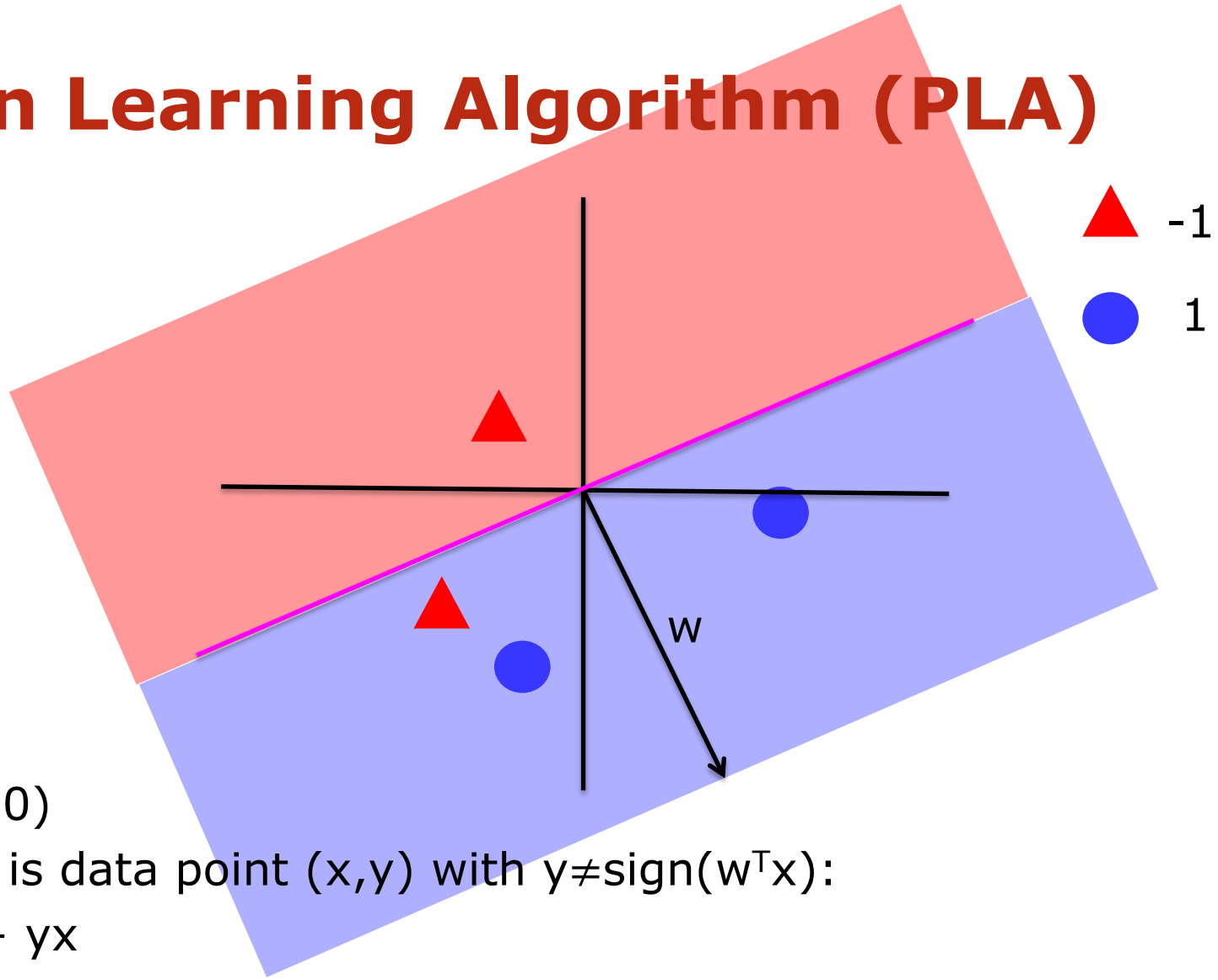
Wrong classification

Make w "more" correct on (x,y)



Perceptron Learning Algorithm (PLA)

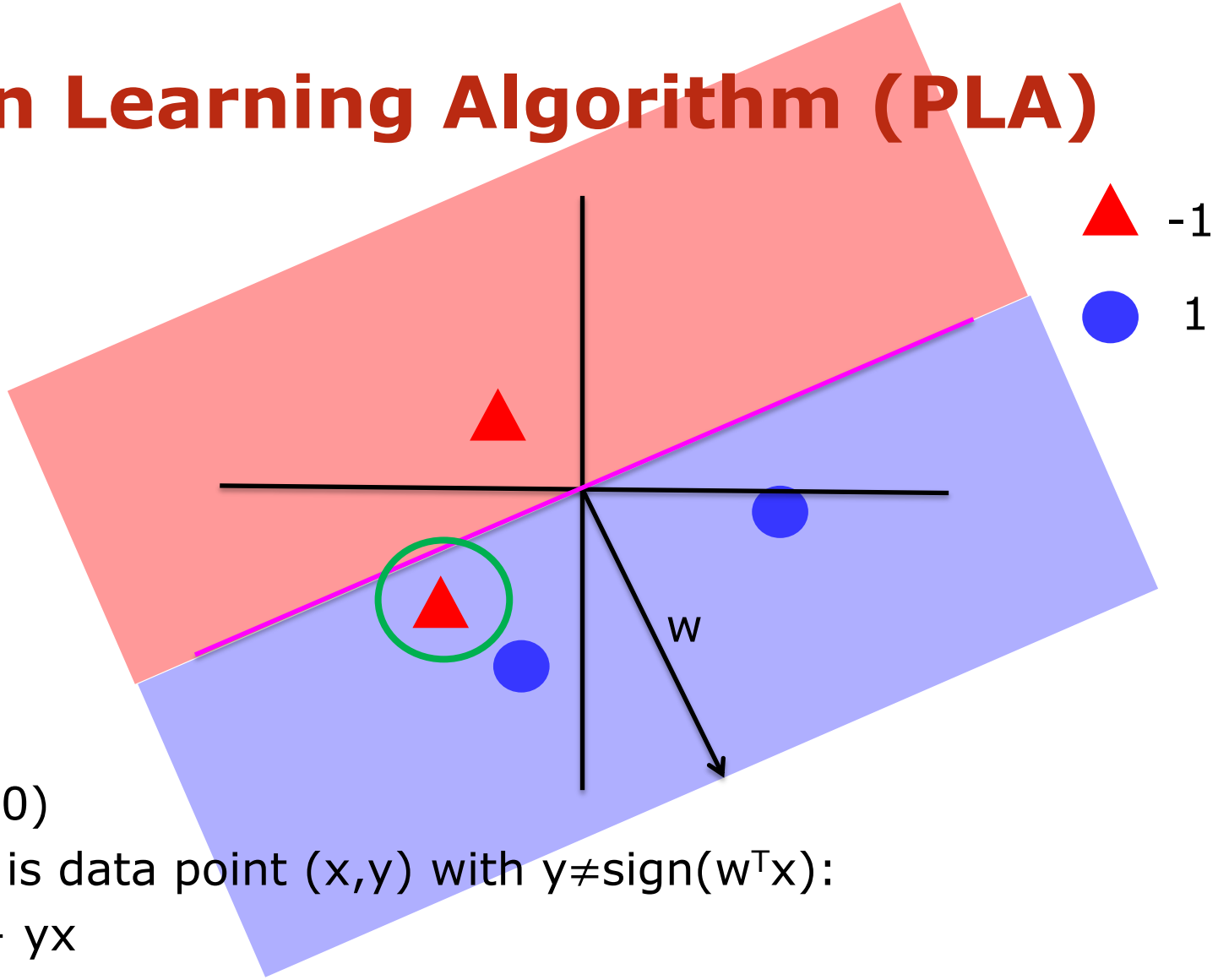
- Example:



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Perceptron Learning Algorithm (PLA)

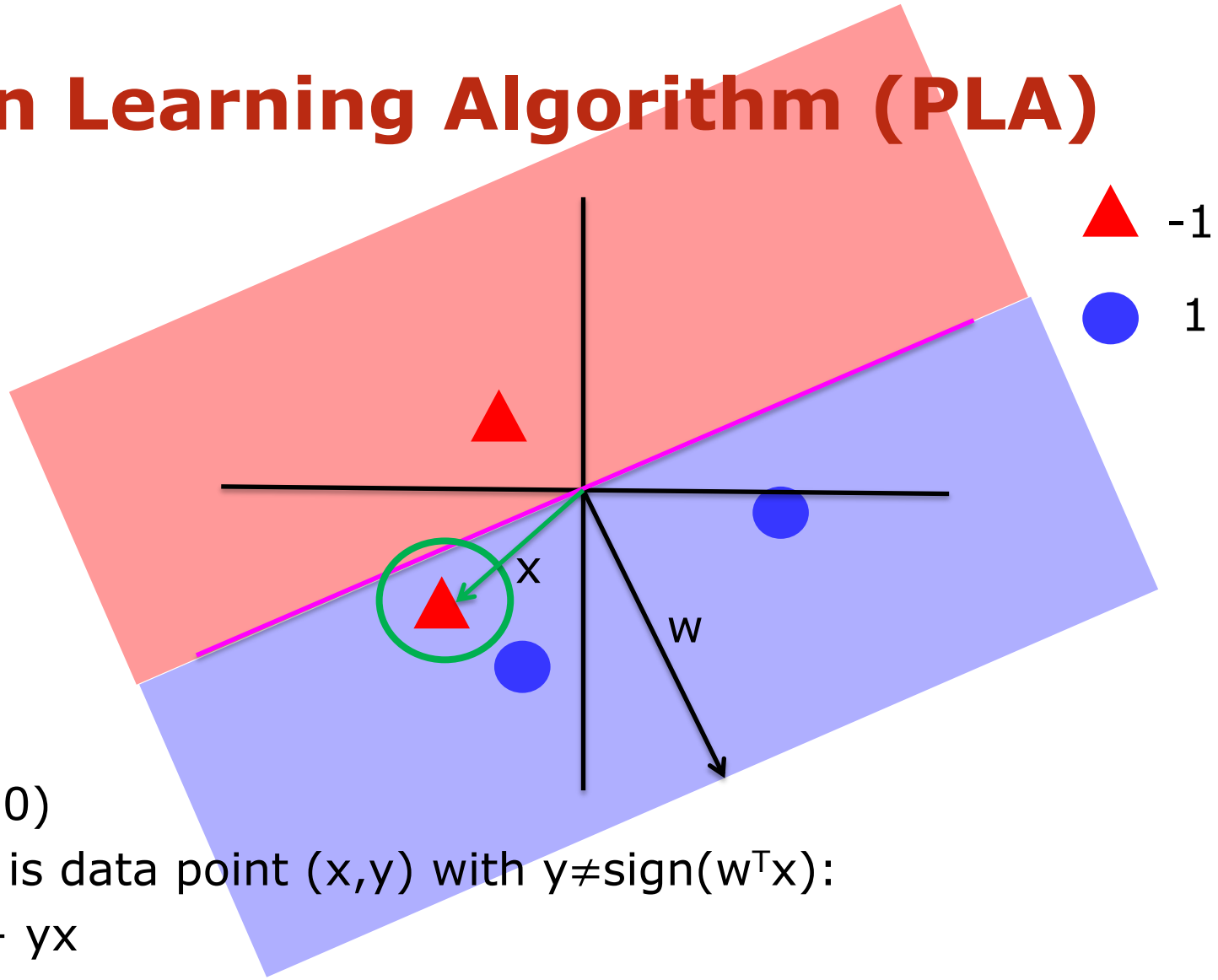
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Perceptron Learning Algorithm (PLA)

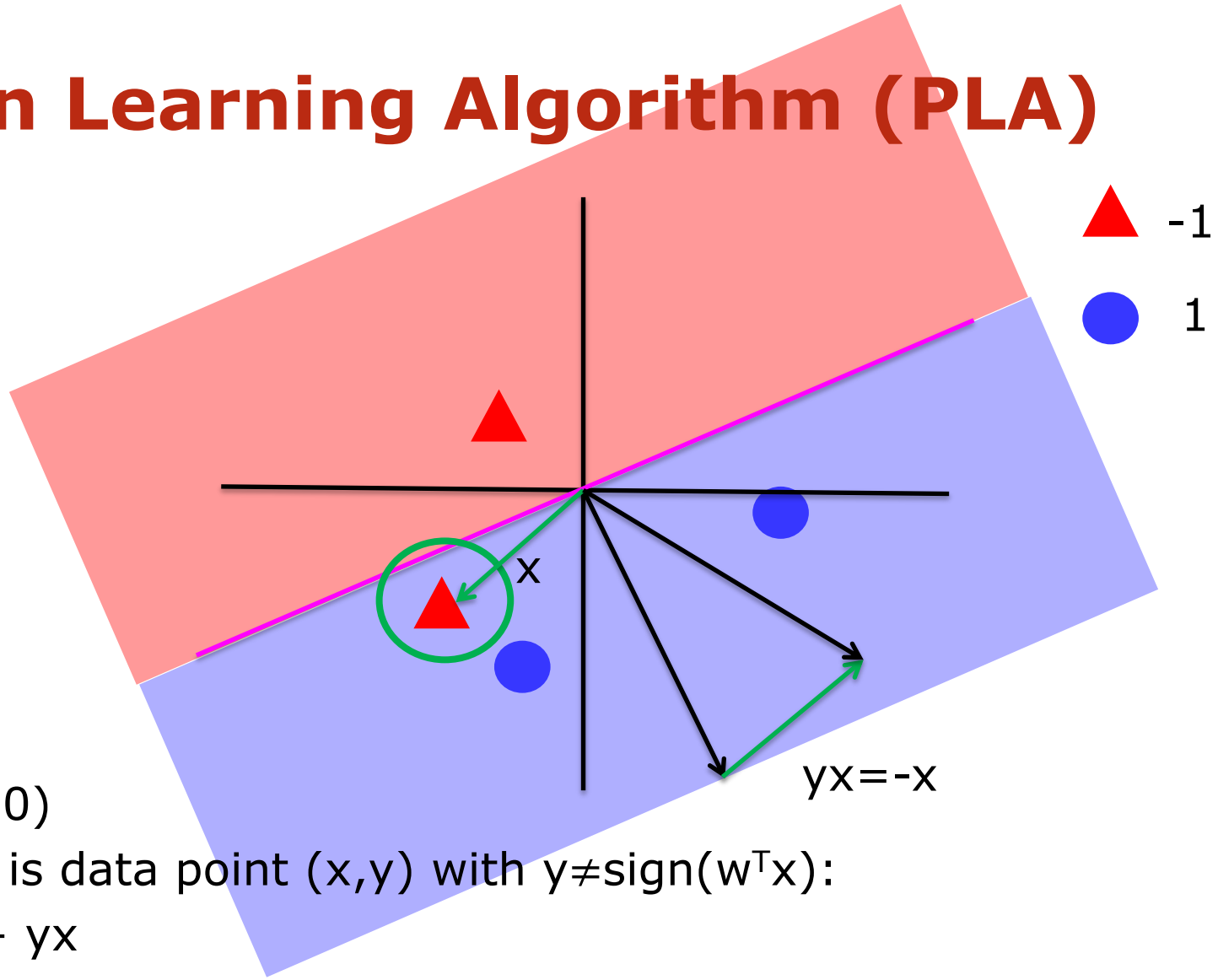
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Perceptron Learning Algorithm (PLA)

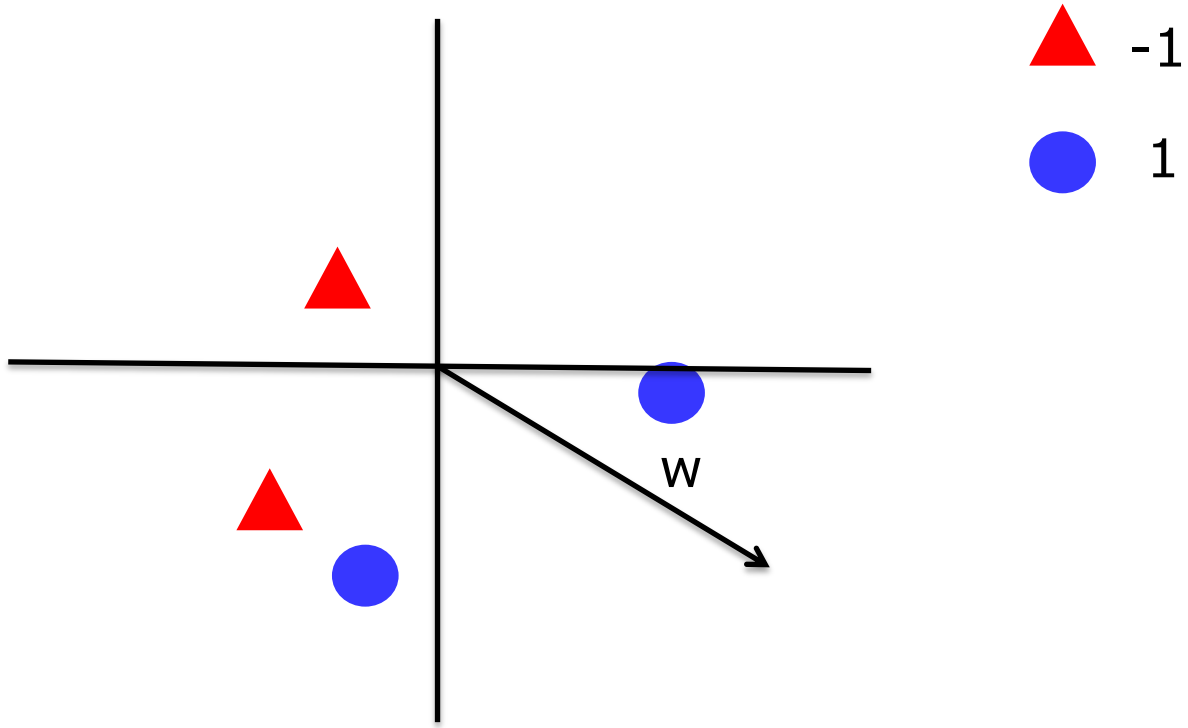
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Perceptron Learning Algorithm (PLA)

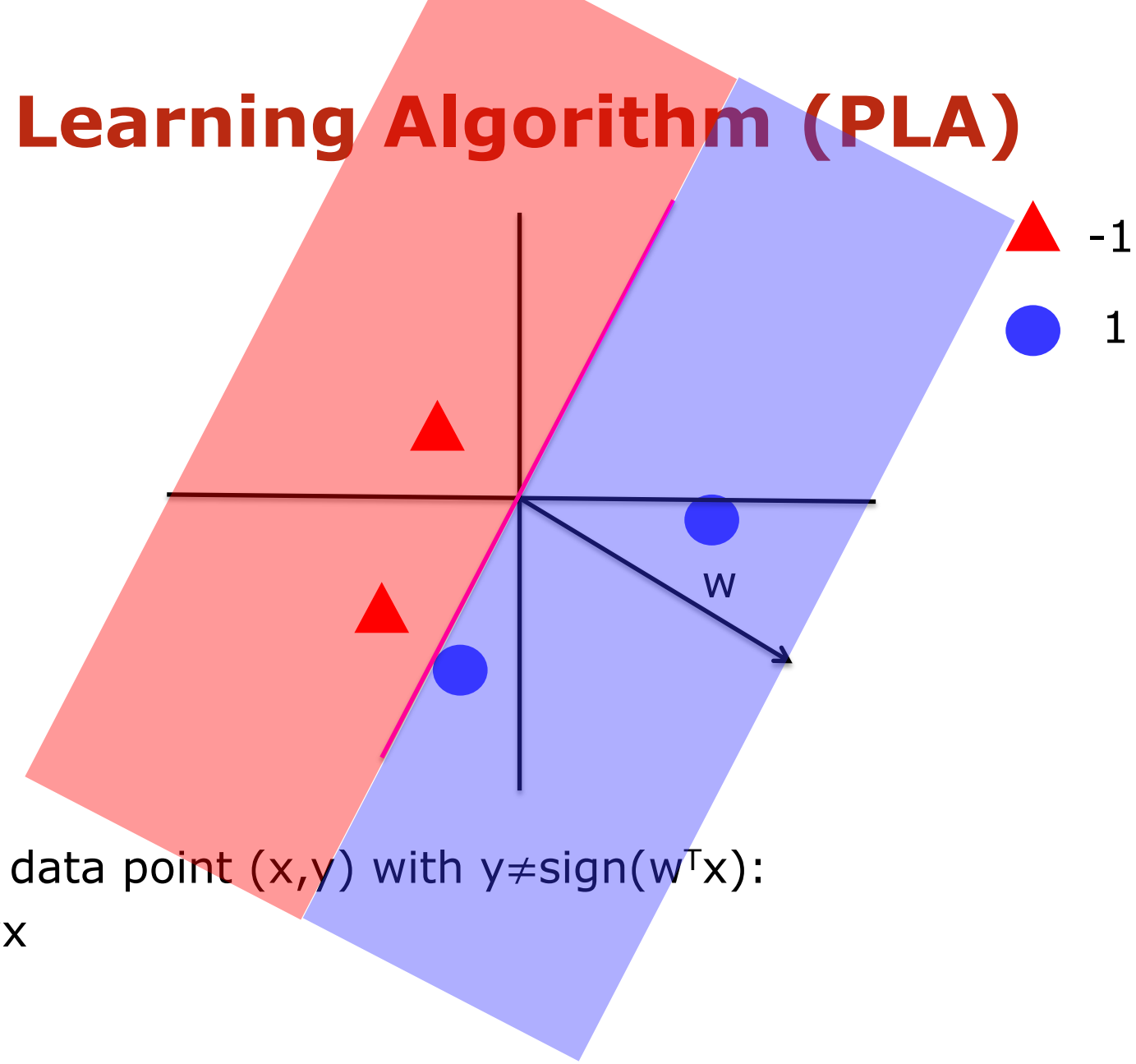
- Example:



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Perceptron Learning Algorithm (PLA)

- Example:



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Perceptron Learning Algorithm (PLA)

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  - w = (0,0,…,0)
  - While there is data point (x,y) with $y \neq \text{sign}(w^T x)$:
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Make w "more" correct on (x,y)

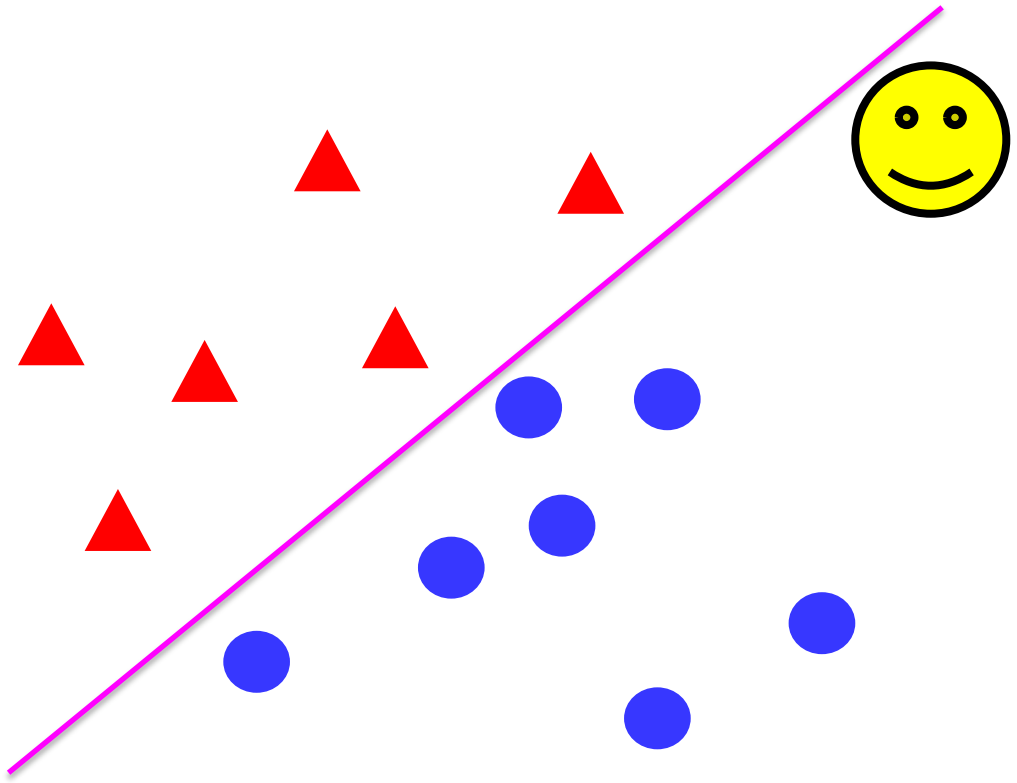
- Observation:
  - May become wrong on other points!

- Is it even a good algorithm??
  - Can prove (exercise in [LFD]): If *linearly separable*, then terminates with perfect classification of training data.
  - Quite non-trivial to prove, not curriculum.



Linearly Separable

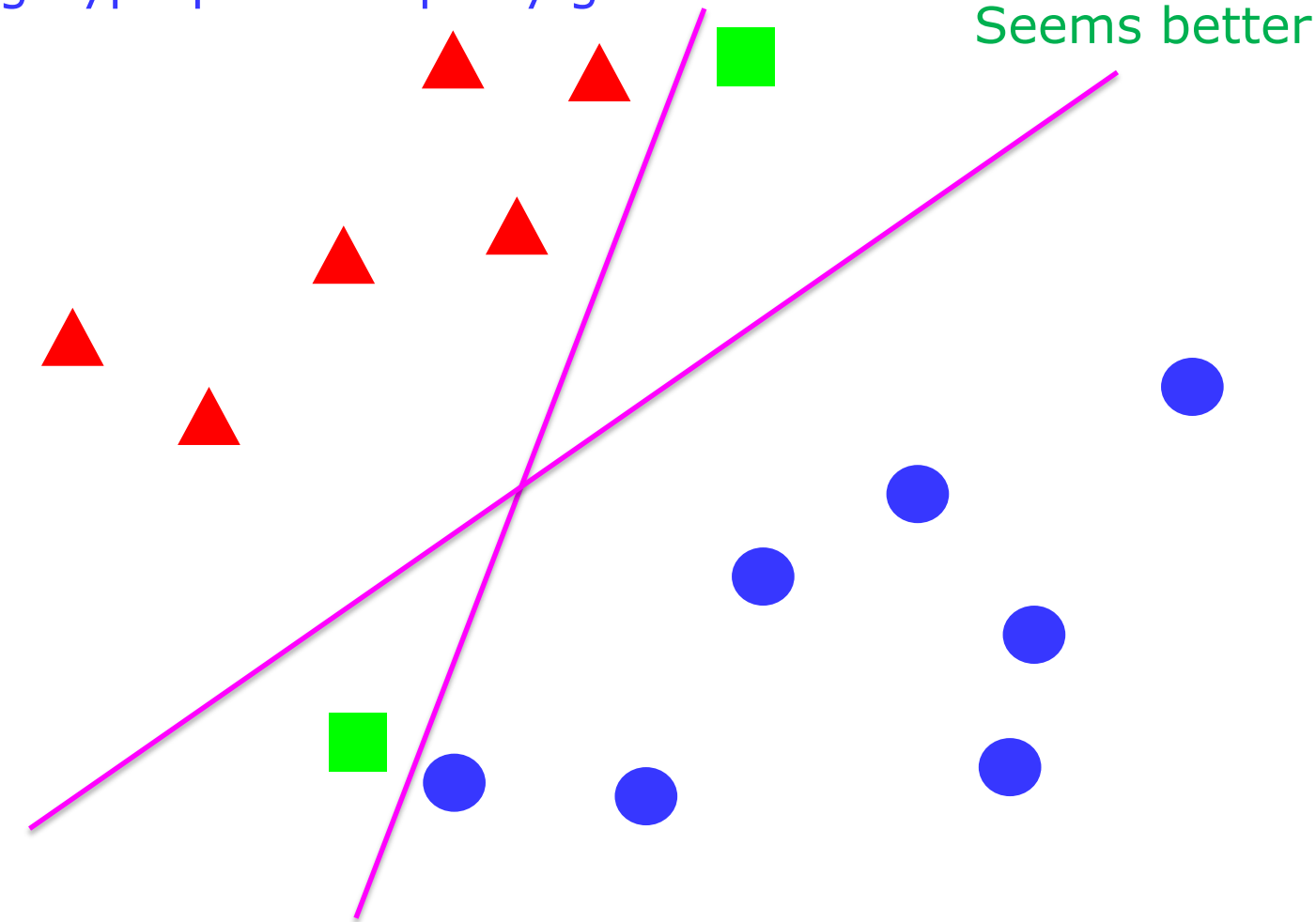
- Definition:
  - Linearly separable if there exists a separating hyperplane (line in 2d).



- Linearly separable!

Linearly Separable

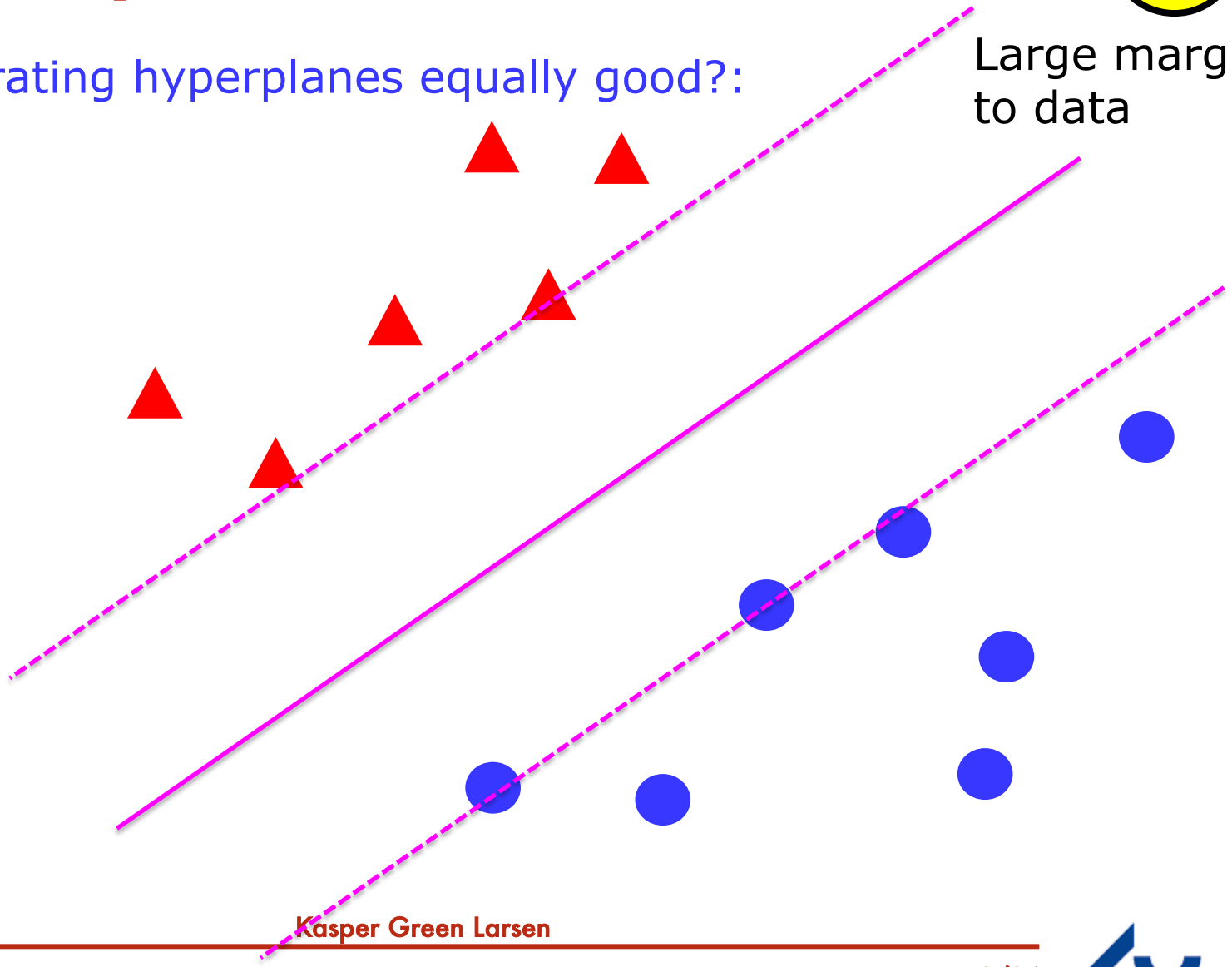
- Are all separating hyperplanes equally good?:



Linearly Separable



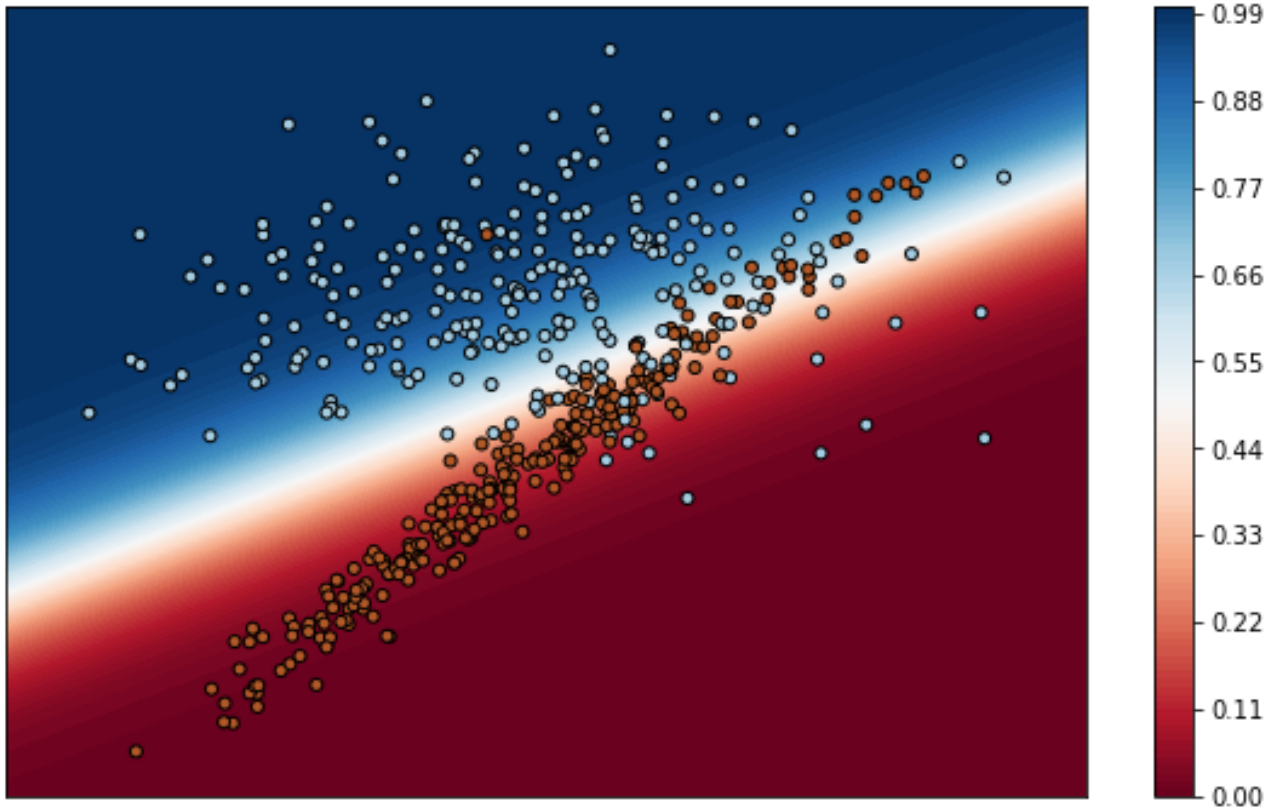
- Are all separating hyperplanes equally good?:





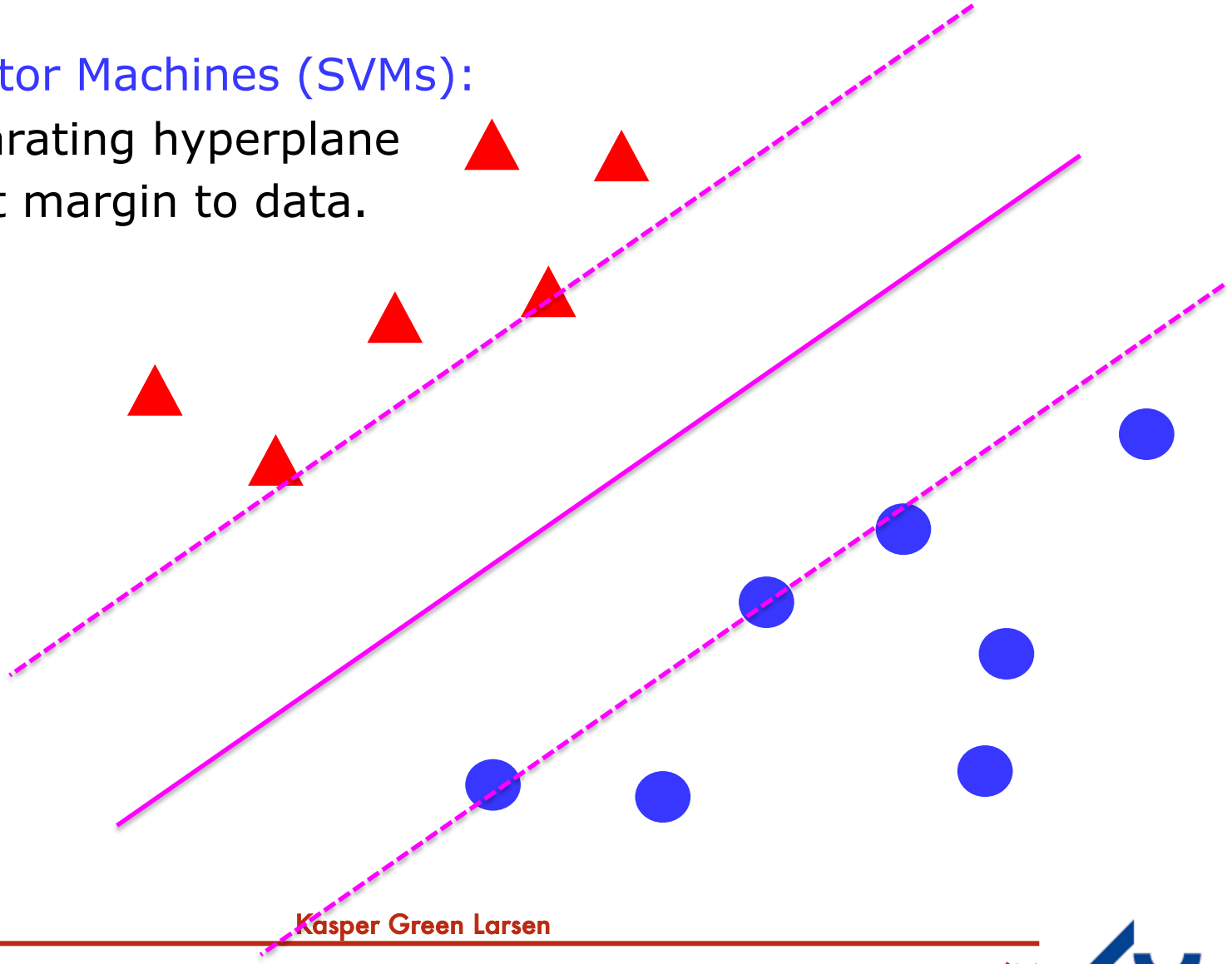
Logistic Regression

- Are all separating hyperplanes equally good?:
- More certain on points far from hyperplane.
- Seems good if very certain (and correct) on all training data.



Support Vector Machines

- Support Vector Machines (SVMs):
  - Use separating hyperplane with largest margin to data.

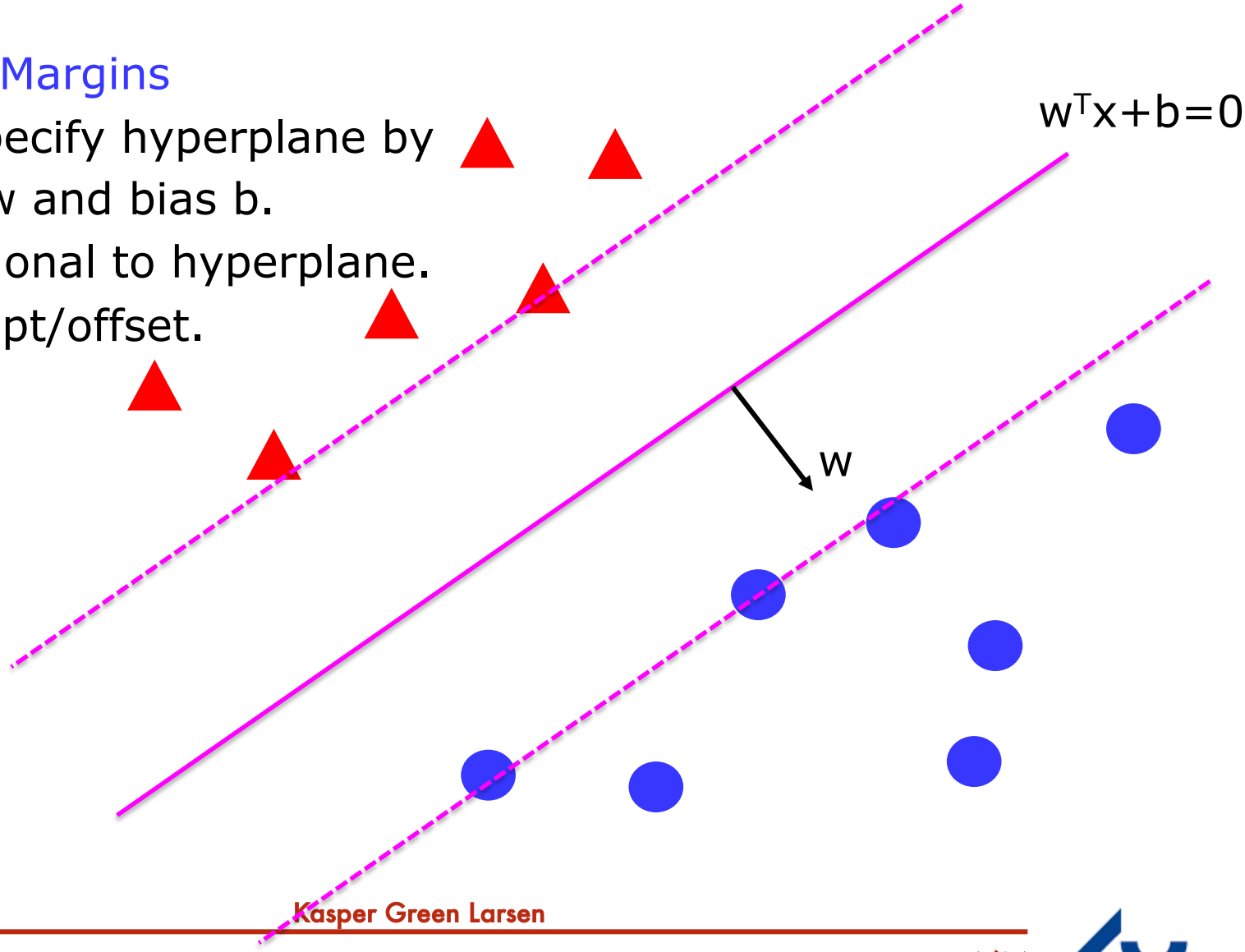


For now:

Assume linearly separable!

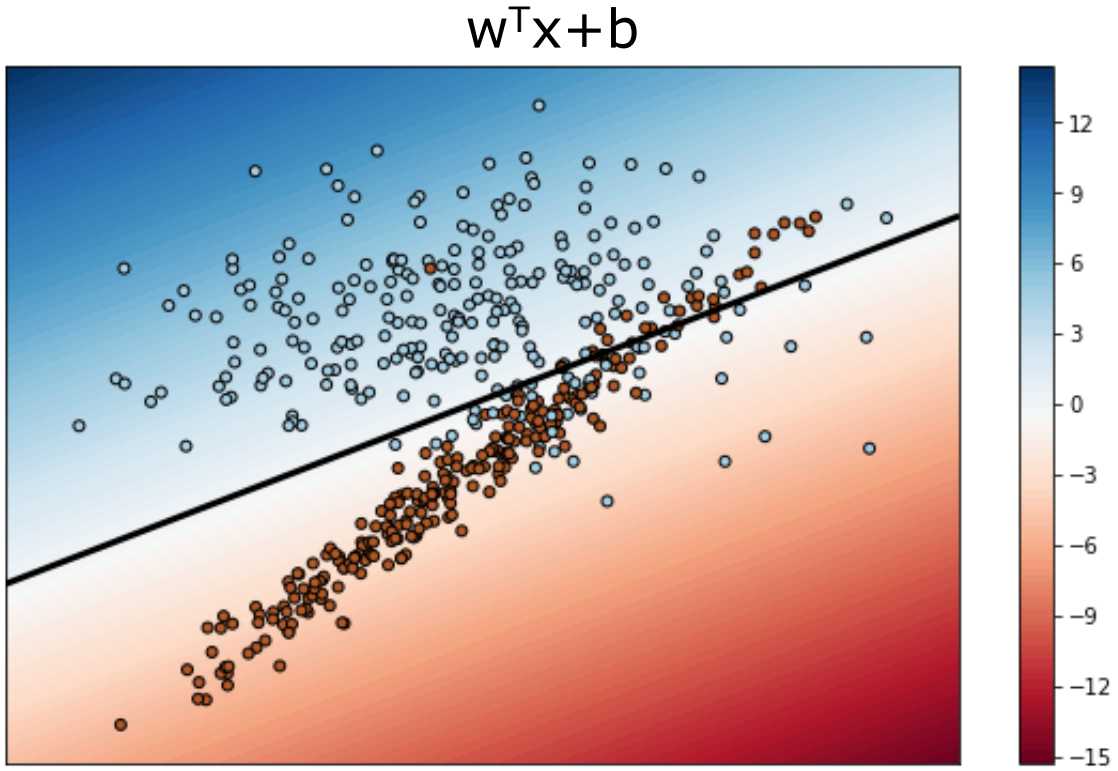
Support Vector Machines

- Formalizing Margins
  - Recall specify hyperplane by vector w and bias b.
  - w orthogonal to hyperplane.
  - b intercept/offset.



Support Vector Machines

- Formalizing Margins
  - $w^T x+b$ signed distance of x from hyperplane (up to scaling).

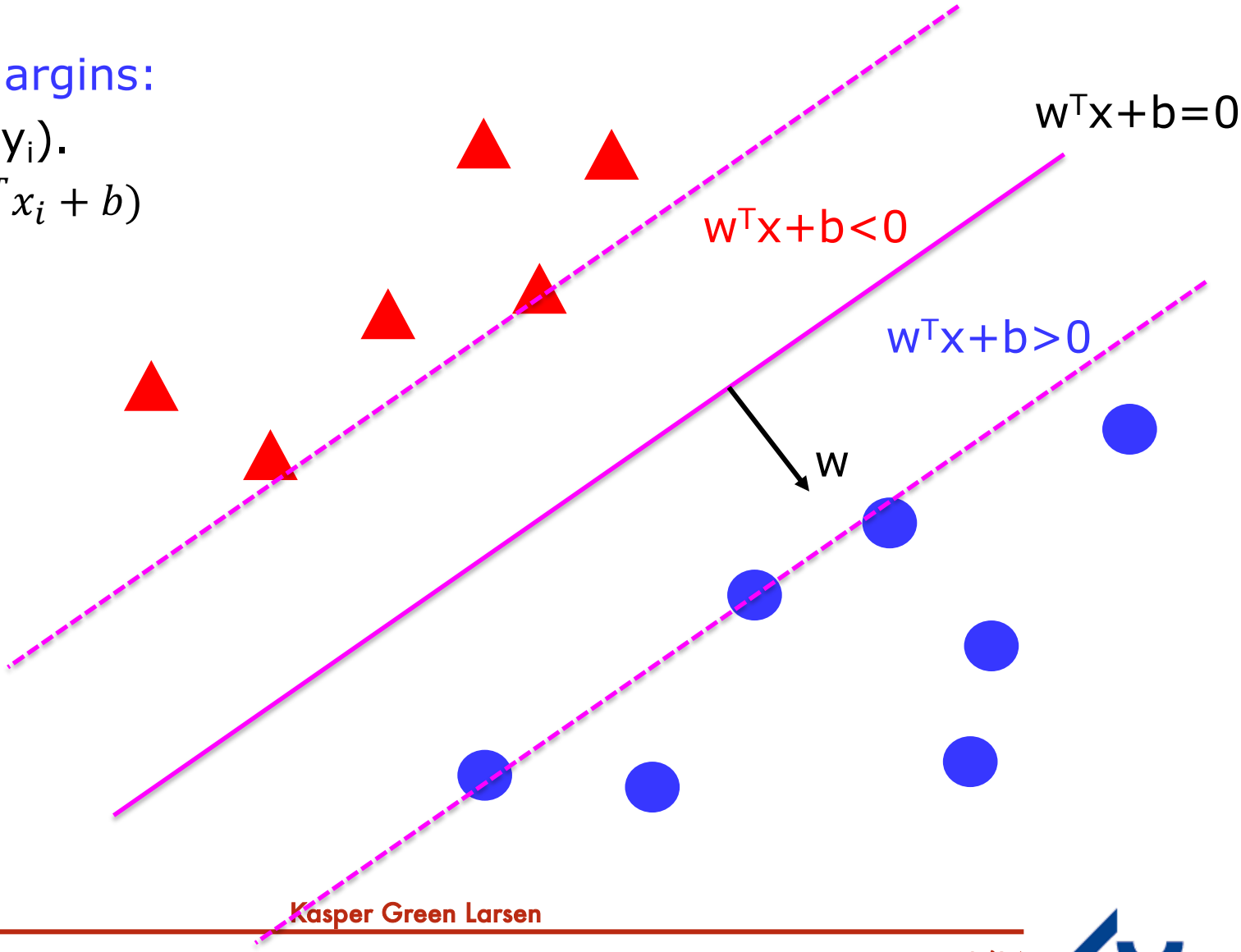


Support Vector Machines

- Functional Margins:
  - Point (x_i, y_i).
  - $\hat{\gamma}_i = y_i(w^T x_i + b)$

$\hat{\gamma}_i$ positive if on correct side of hyperplane.

$\hat{\gamma}_i$ negative if on wrong side of hyperplane.

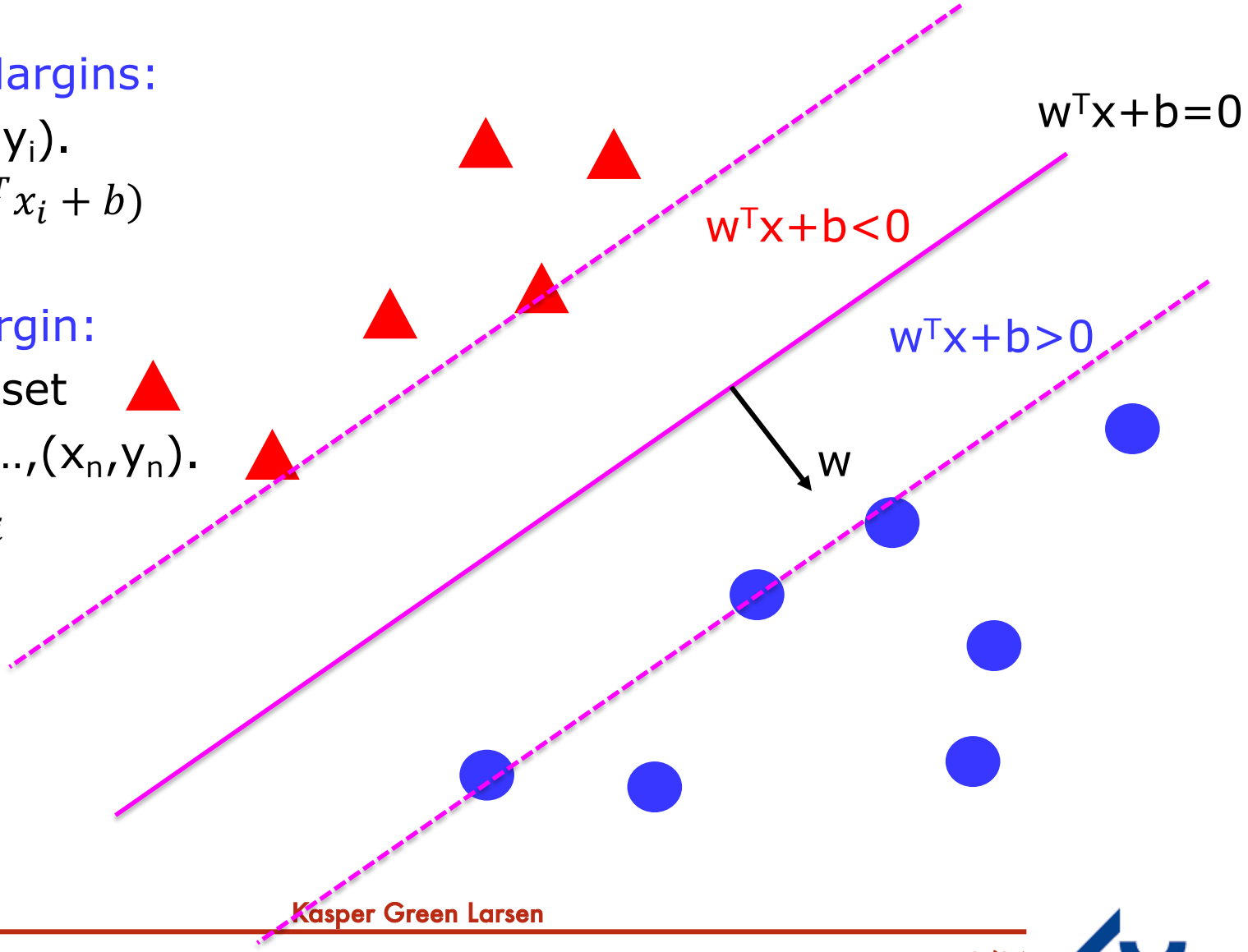


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Support Vector Machines

- Functional Margins:
  - Point (x_i,y_i).
  - $\hat{\gamma}_i = y_i(w^T x_i + b)$
- Function Margin:
  - Training set $(x_1,y_1),\ldots,(x_n,y_n)$.
  - $\hat{\gamma} = \min_i \hat{\gamma}_i$



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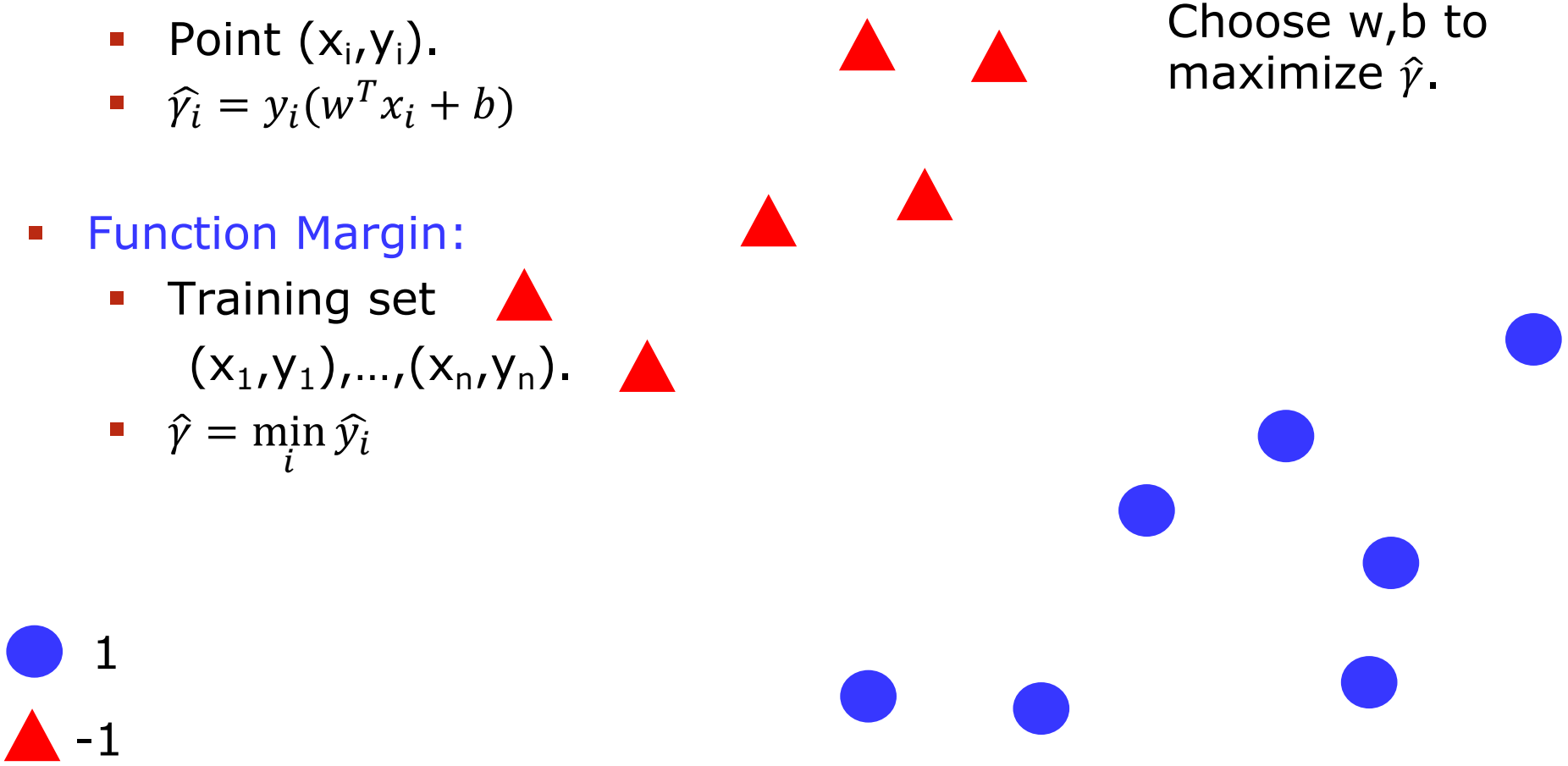
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Support Vector Machines

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SVM first approach:

Choose w,b to maximize $\hat{\gamma}$.

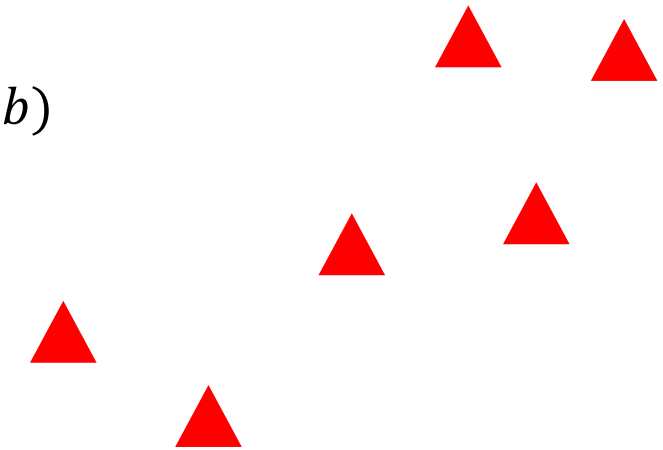


Support Vector Machines

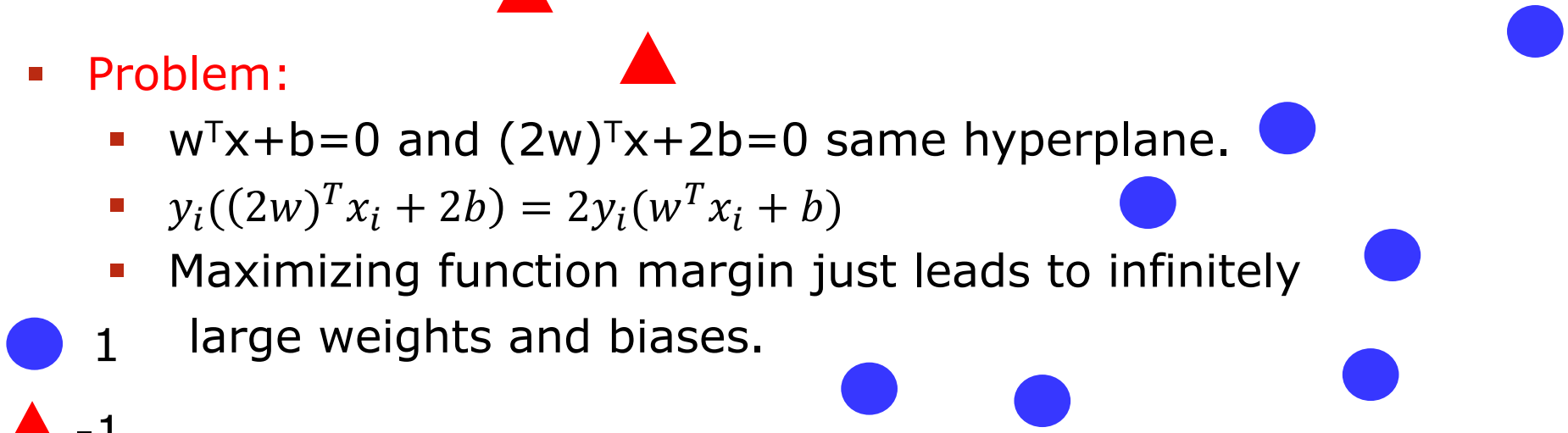
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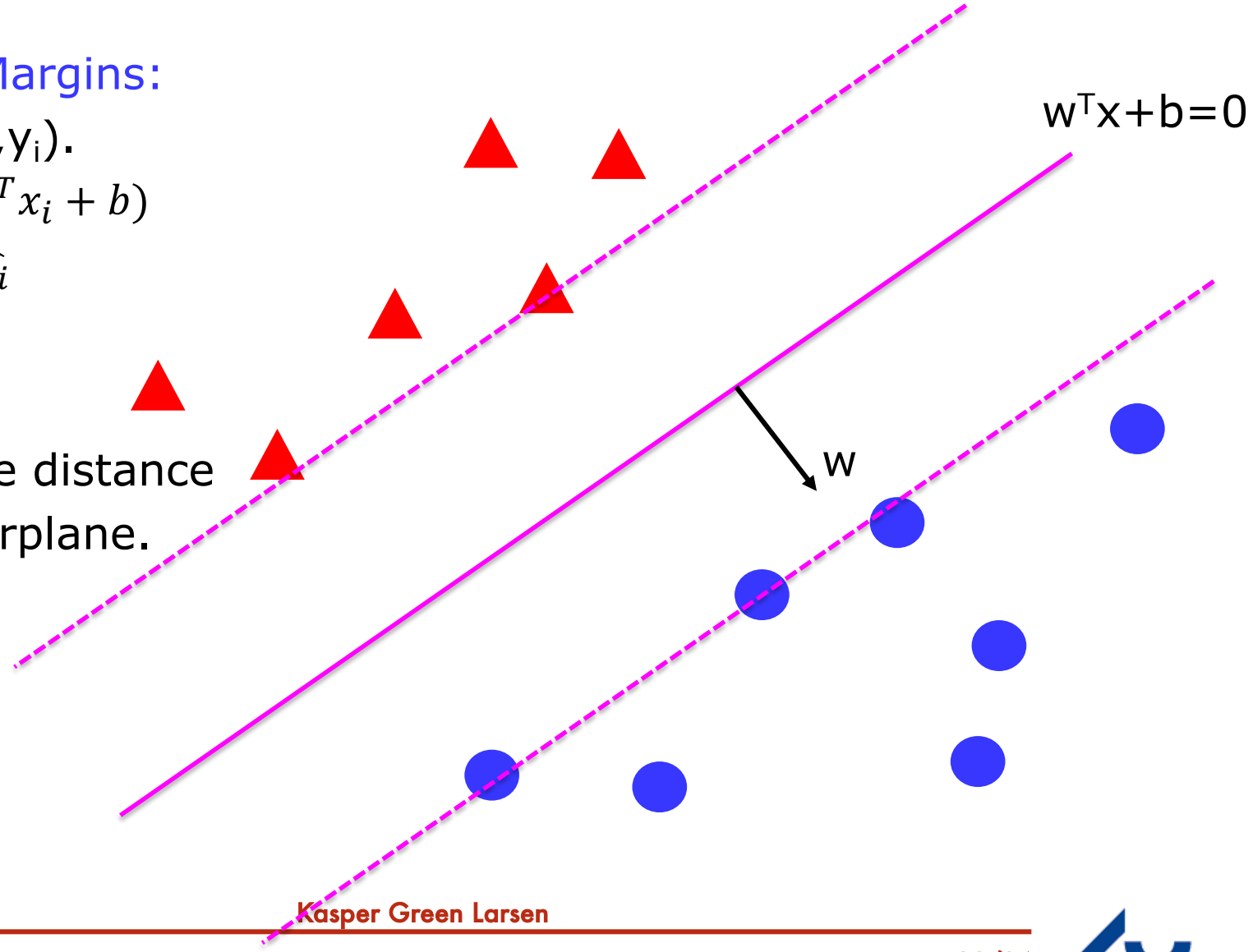
- Problem:
  - $w^T x+b=0$ and $(2w)^T x+2b=0$ same hyperplane.
  - $y_i((2w)^T x_i + 2b) = 2y_i(w^T x_i + b)$
  - Maximizing function margin just leads to infinitely large weights and biases.



Support Vector Machines

- Functional Margins:
  - Point (x_i, y_i).
  - $\hat{\gamma}_i = y_i(w^T x_i + b)$
  - $\hat{\gamma} = \min_i \hat{\gamma}_i$

- Really after:
  - Maximize distance to hyperplane.

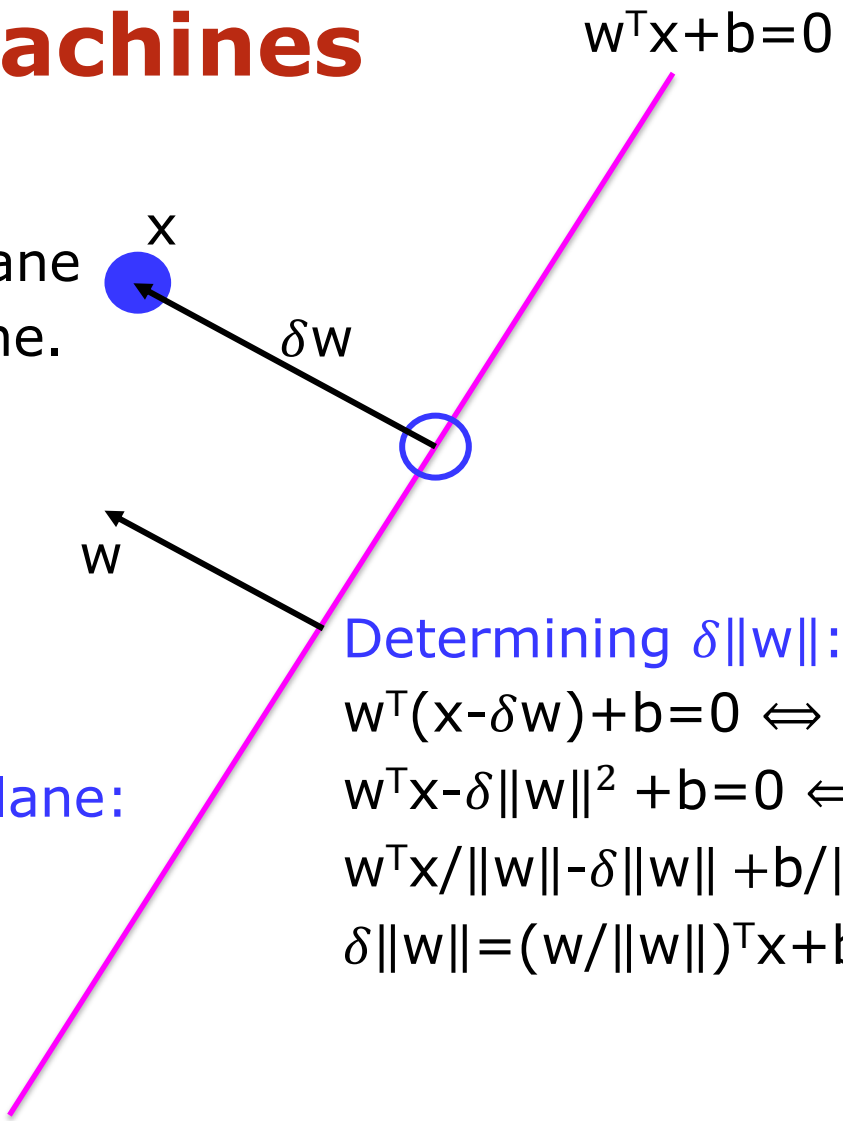


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Support Vector Machines

- Distance to hyperplane
  - Closest point on hyperplane is projection onto hyperplane.
- Projection of x:
  - x-δw for some δ.
- Must be on hyperplane:
  - $w^T(x-\delta w)+b=0$
- (Signed) Distance to hyperplane:
  - $\delta\|w\|$
  - $(w/\|w\|)^Tx+b/\|w\|$.



Determining $\delta\|w\|$:

$w^T(x-\delta w)+b=0 \Leftrightarrow$

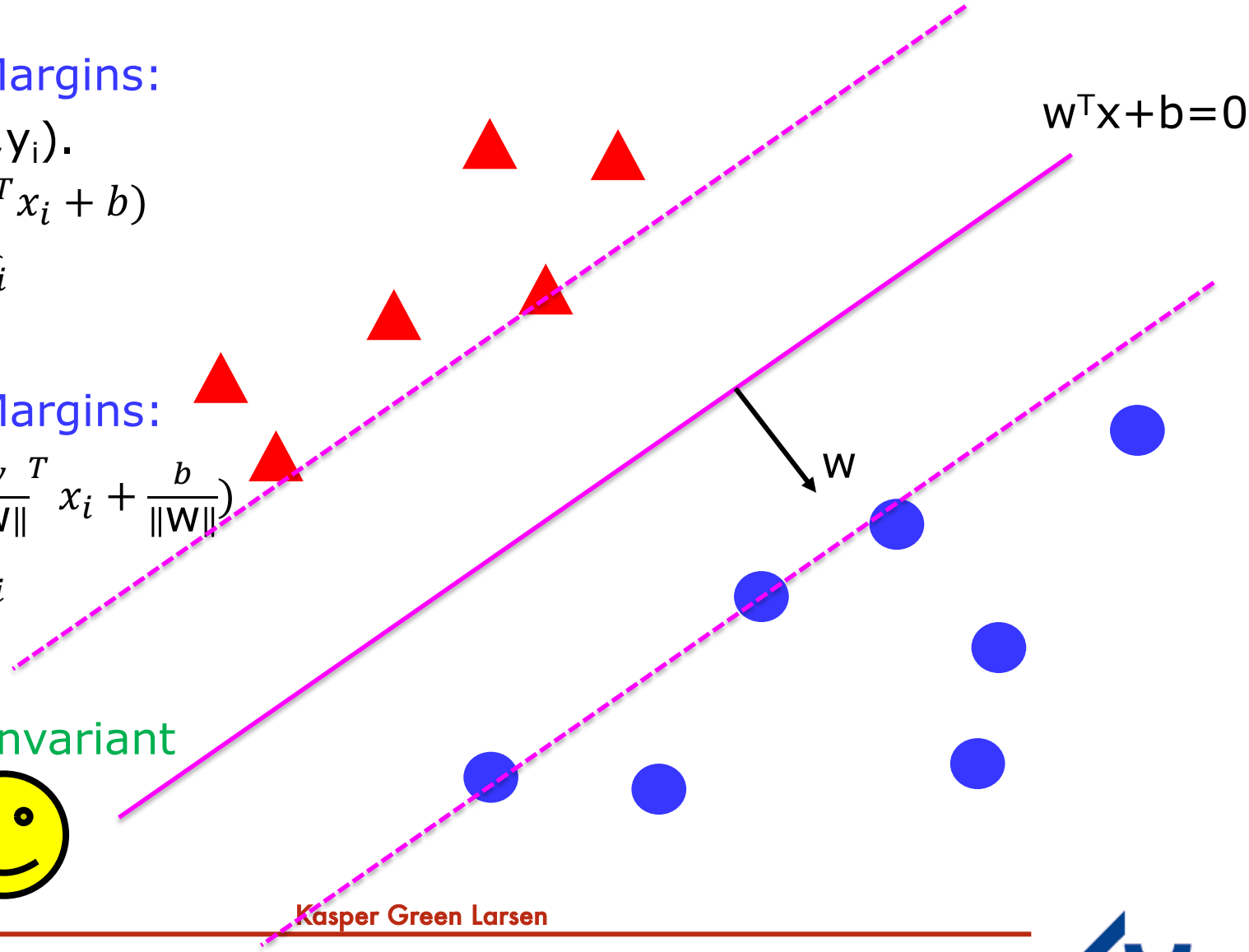
$w^Tx-\delta\|w\|^2+b=0 \Leftrightarrow$

$w^Tx/\|w\|-\delta\|w\|+b/\|w\|=0 \Leftrightarrow$

$\delta\|w\|=(w/\|w\|)^Tx+b/\|w\|$

Support Vector Machines

- Functional Margins:
  - Point (x_i, y_i).
  - $\hat{\gamma}_i = y_i(w^T x_i + b)$
  - $\hat{\gamma} = \min_i \hat{y}_i$
- Geometric Margins:
  - $\gamma_i = y_i(\frac{w}{\|w\|}^T x_i + \frac{b}{\|w\|})$
  - $\gamma = \min_i \gamma_i$



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Scale-invariant



Support Vector Machines

- Functional Margins:
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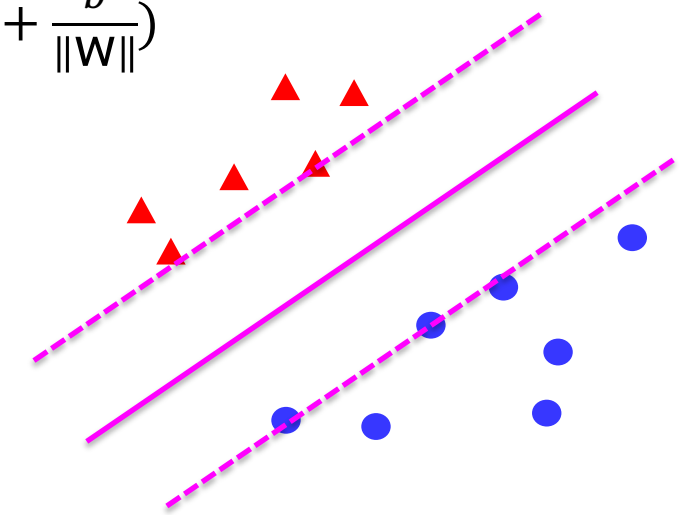
Force functional and geometric equal

- Support Vector Machines
- Solve constrained optim. problem:
  - $\max_{w,b,\gamma} \gamma$

subject to:

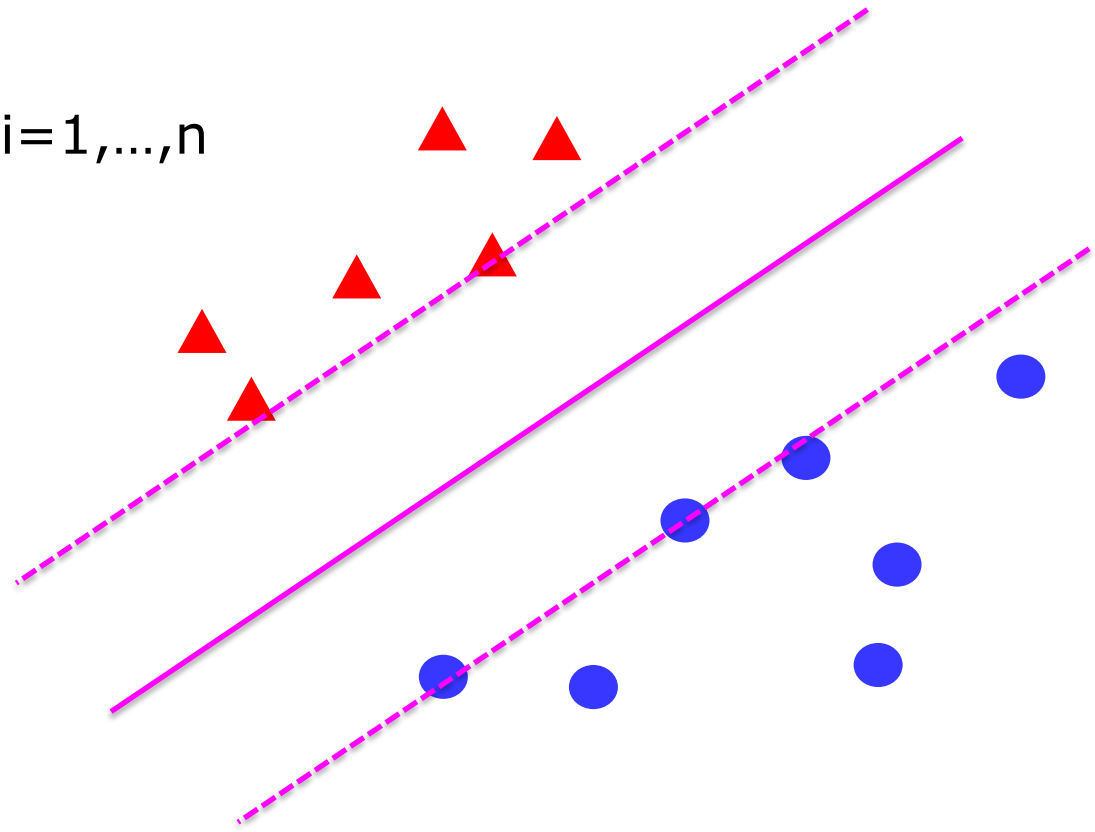
- $y_i(w^T x_i + b) \geq \gamma$ i=1,…,n
- $\|w\|=1$

Maximize functional margin



Support Vector Machines

- Support Vector Machines
  - $\max_{w,b,\gamma} \gamma$
  - subject to:
    - $y_i(w^T x_i + b) \geq \gamma$ i=1,…,n
    - ||w||=1



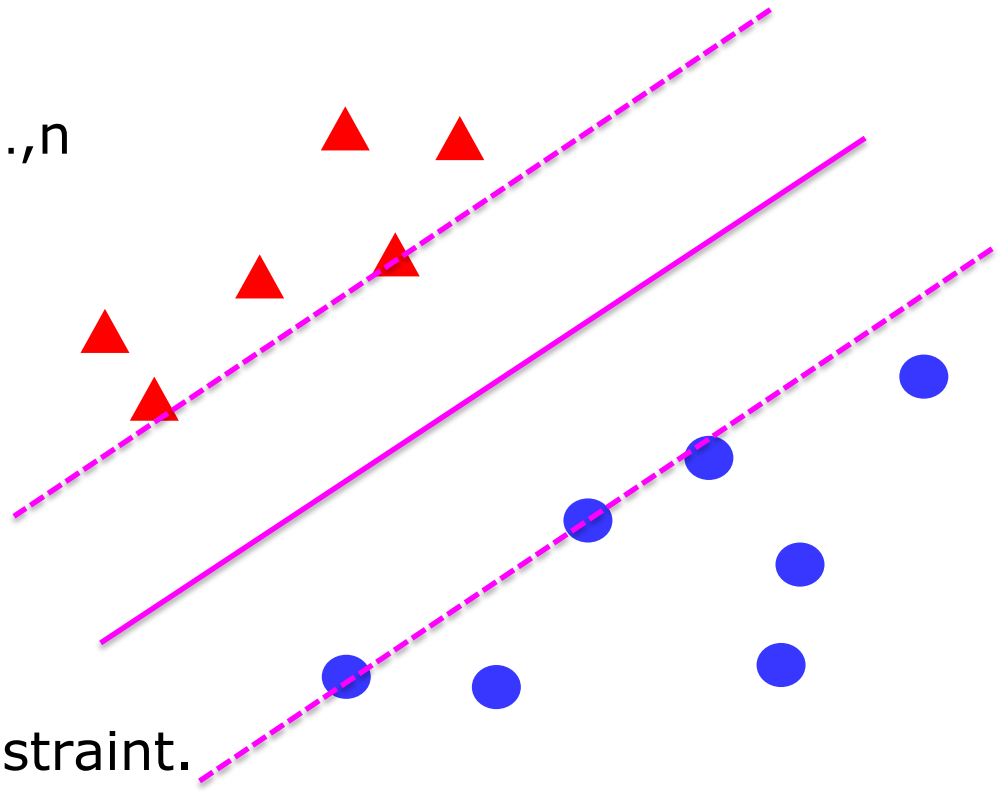
Support Vector Machines

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Rewriting Problem

- Support Vector Machines
  - $\max_{w,b,\gamma} \gamma$
  - subject to:
    - $y_i(w^T x_i + b) \geq \gamma$ i=1,…,n
    - ||w||=1



- If we ignore ||w||=1, then a linear program and can be solved efficiently using standard software.

- ||w||=1 nasty (non-convex) constraint.
- Let us rewrite problem.

Rewriting Problem

Recall geometric margin:

$$\gamma_i = y_i\left(\frac{w^T}{\|w\|}x_i + \frac{b}{\|w\|}\right)$$

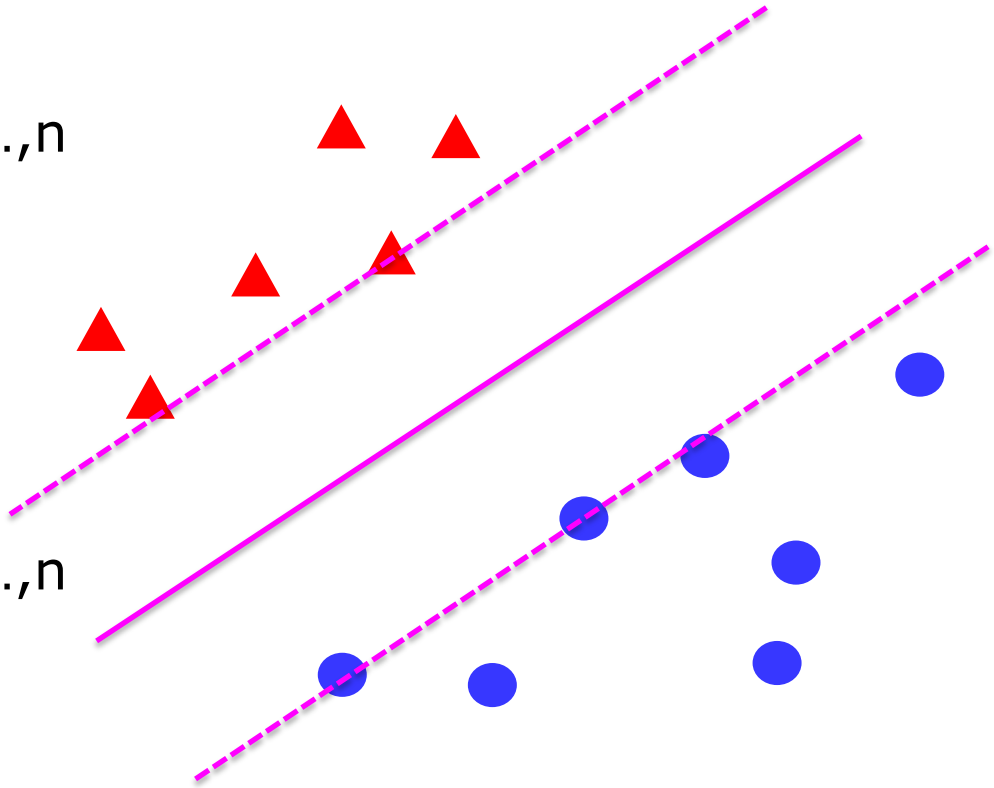
$$\gamma_i = \frac{\hat{\gamma}_i}{\|w\|}$$

- Support Vector Machines
  - $\max_{w,b,\gamma} \gamma$
  - subject to:
    - $y_i(w^T x_i + b) \geq \gamma$ i=1,…,n
    - $\|w\|=1$

- $\max_{w,b,\hat{\gamma}} \frac{\hat{\gamma}}{\|w\|}$

Maximize geometric margin

- subject to:
  - $y_i(w^T x_i + b) \geq \hat{\gamma}$ i=1,…,n



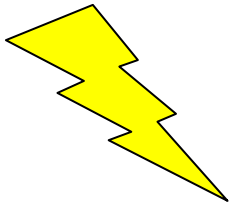
Rewriting Problem

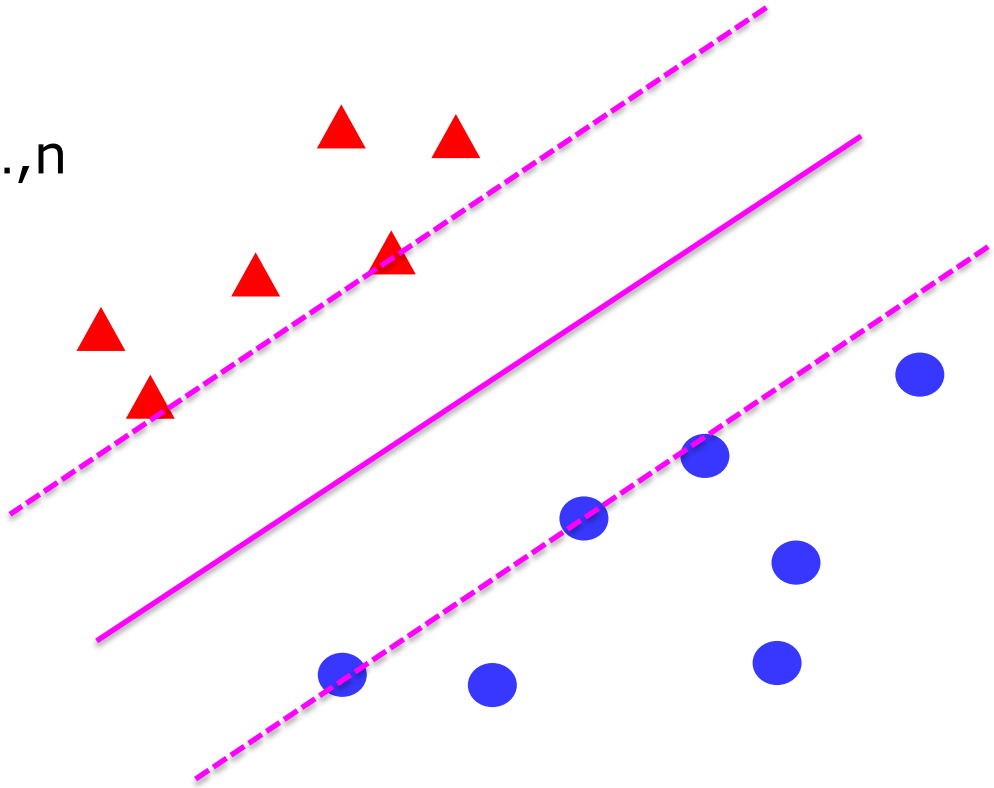
- Support Vector Machines
  - $\max_{w,b,\hat{\gamma}} \frac{\hat{\gamma}}{\|w\|}$
  - subject to:
    - $y_i(w^T x_i + b) \geq \hat{\gamma}$ i=1,…,n

Only linear constraints



Objective function $\frac{\hat{\gamma}}{\|w\|}$ not convex

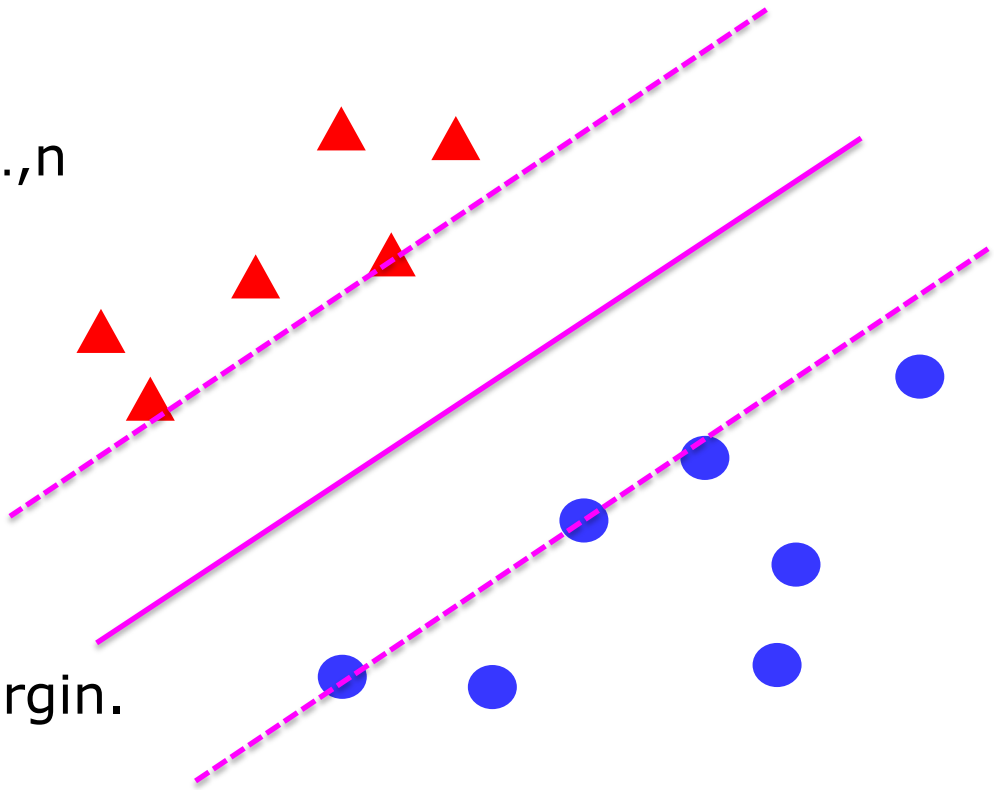






Rewriting Problem

- Support Vector Machines
  - $\max_{w,b,\hat{\gamma}} \frac{\hat{\gamma}}{\|w\|}$
  - subject to:
    - $y_i(w^T x_i + b) \geq \hat{\gamma}$ i=1,…,n



- Observation:
  - If $w, b, \hat{\gamma}$ is optimal solution. then $cw, cb, c\hat{\gamma}$ also optimal.

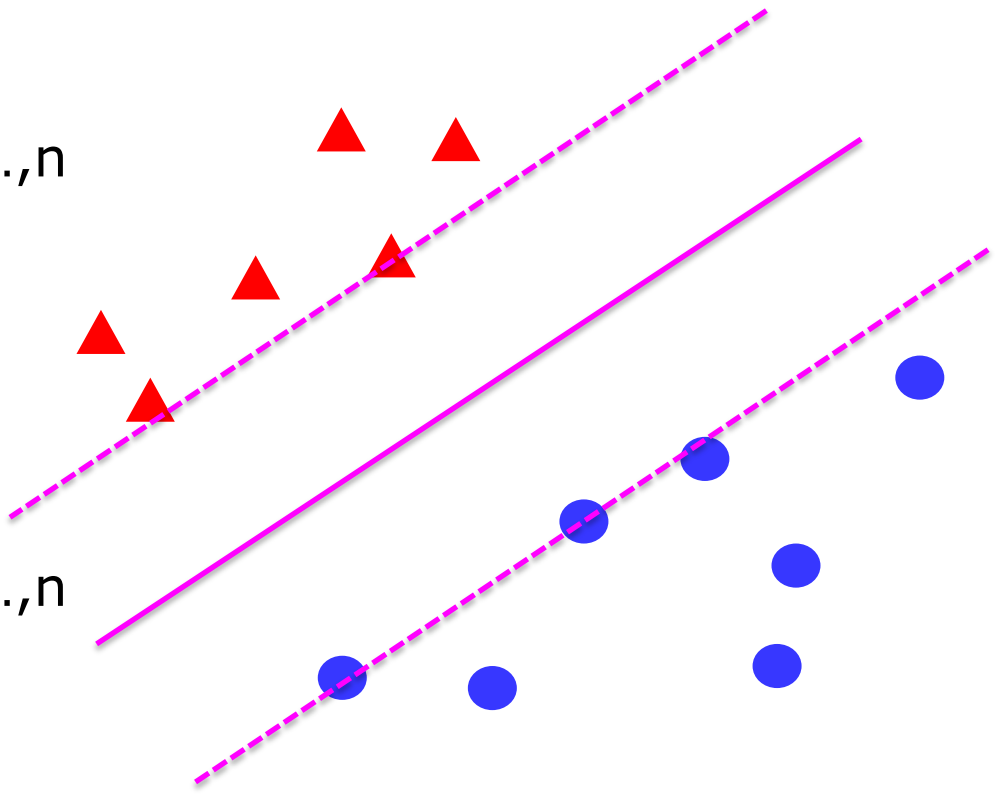
- Key Idea:
  - Force specific functional margin.
  - Constrain $\hat{\gamma} = 1$.
  - Only effect is it forces a specific scaling of w and b.



Rewriting Problem

- Support Vector Machines
  - $\max_{w,b,\hat{\gamma}} \frac{\hat{\gamma}}{\|w\|}$
  - subject to:
    - $y_i(w^T x_i + b) \geq \hat{\gamma}$ i=1,…,n

- Support Vector Machines
  - $\max_{w,b} \frac{1}{\|w\|}$
  - subject to:
    - $y_i(w^T x_i + b) \geq 1$ i=1,…,n



Rewriting Problem

- Support Vector Machines
  - $\max_{w,b,\hat{\gamma}} \frac{\hat{\gamma}}{\|w\|}$
  - subject to:
    - $y_i(w^T x_i + b) \geq \hat{\gamma}$ i=1,…,n
- Support Vector Machines
  - $\max_{w,b} \frac{1}{\|w\|}$
  - subject to:
    - $y_i(w^T x_i + b) \geq 1$ i=1,…,n

- Claim same solution
  - Any hyperplane specifiable as $(cw)^T x+cb$
  - Cost in second formulation is $\frac{1}{\|cw\|} = \frac{1}{c\|w\|}$
  - For second formulation, best choice of c is such that

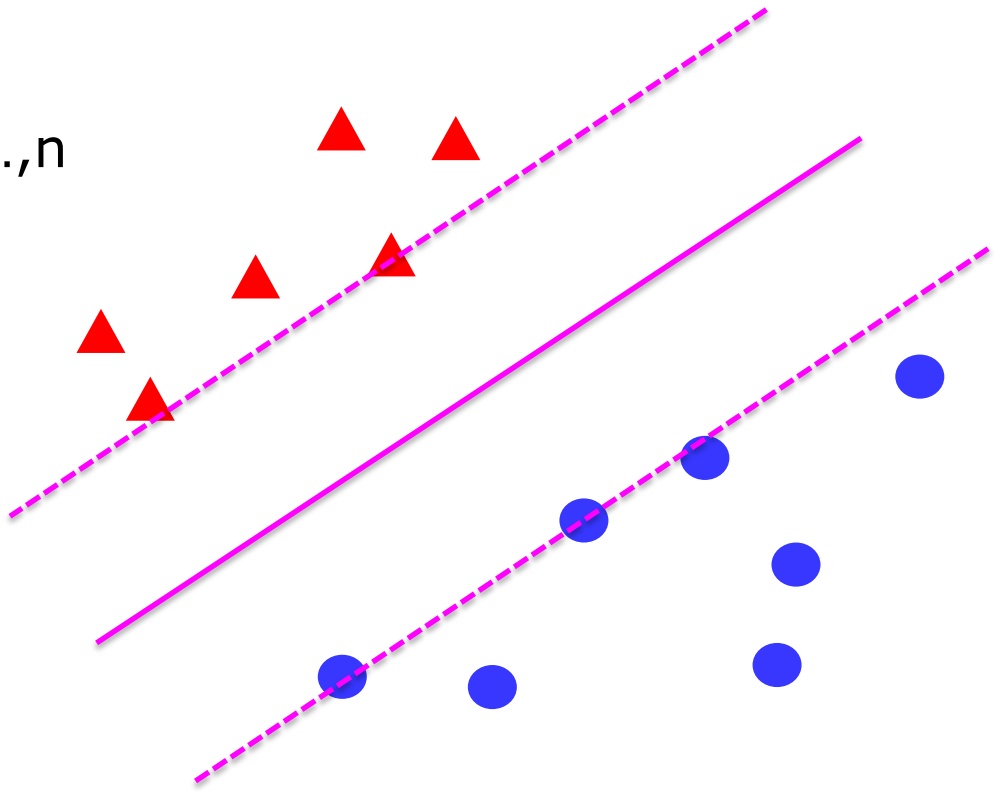
    $\min_i cy_i(w^T x_i + b) = 1$
  - $c=1/\hat{\gamma}$ (of w and b)
  - Cost in second formulation is $\frac{1}{c\|w\|} = \frac{\hat{\gamma}}{\|w\|}$

Geometric margin



Rewriting Problem

- Support Vector Machines
  - $\max_{w,b} \frac{1}{\|w\|}$
  - subject to:
    - $y_i(w^T x_i + b) \geq 1$ i=1,…,n
- Still not convex objective
- Observation:
  - Maximizing $\frac{1}{\|w\|}$ same as
  - Minimizing $\|w\|$, same as
  - Minimizing $\frac{1}{2}\|w\|^2$





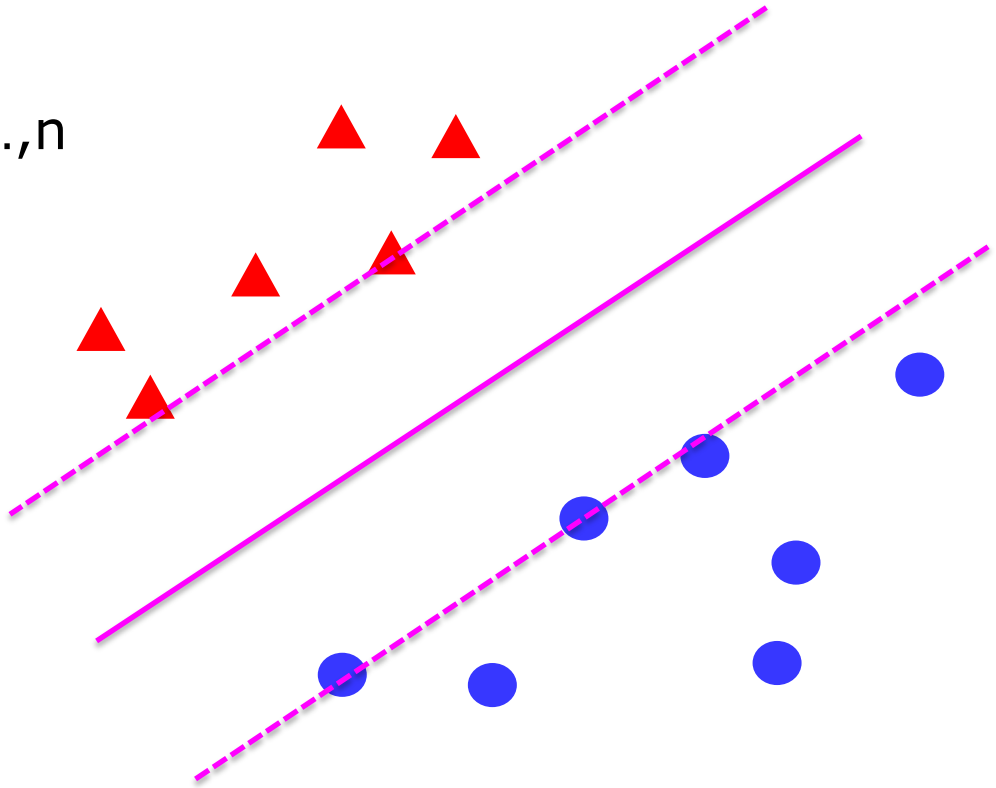
½ will be convenient later

Convex function of w

Rewriting Problem

- Support Vector Machines
  - $\min_{w,b} \frac{1}{2}\|w\|^2$
  - subject to:
    - $y_i(w^T x_i + b) \geq 1$ i=1,…,n

- There are efficient algorithms for convex optimization under linear constraints.
- Similar to convex being ”easy” for gradient descent.



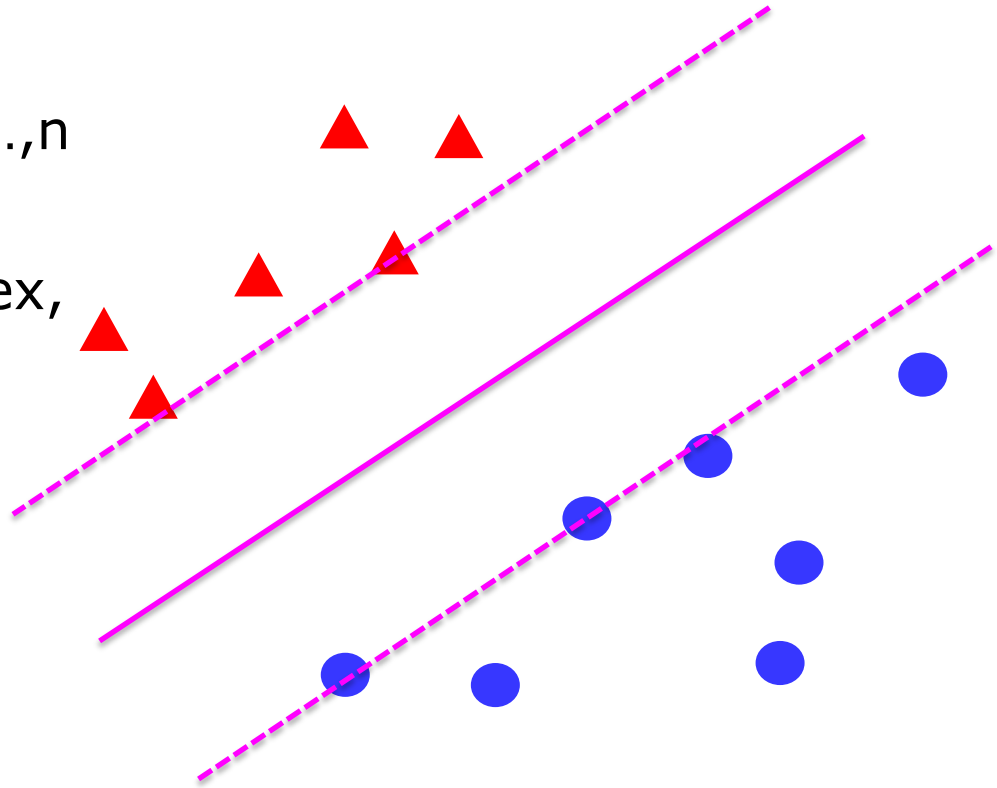
We like convex



Rewriting Problem

- Support Vector Machines
  - $\min_{w,b} \frac{1}{2}\|w\|^2$
  - subject to:
    - $y_i(w^T x_i + b) \geq 1$ i=1,…,n

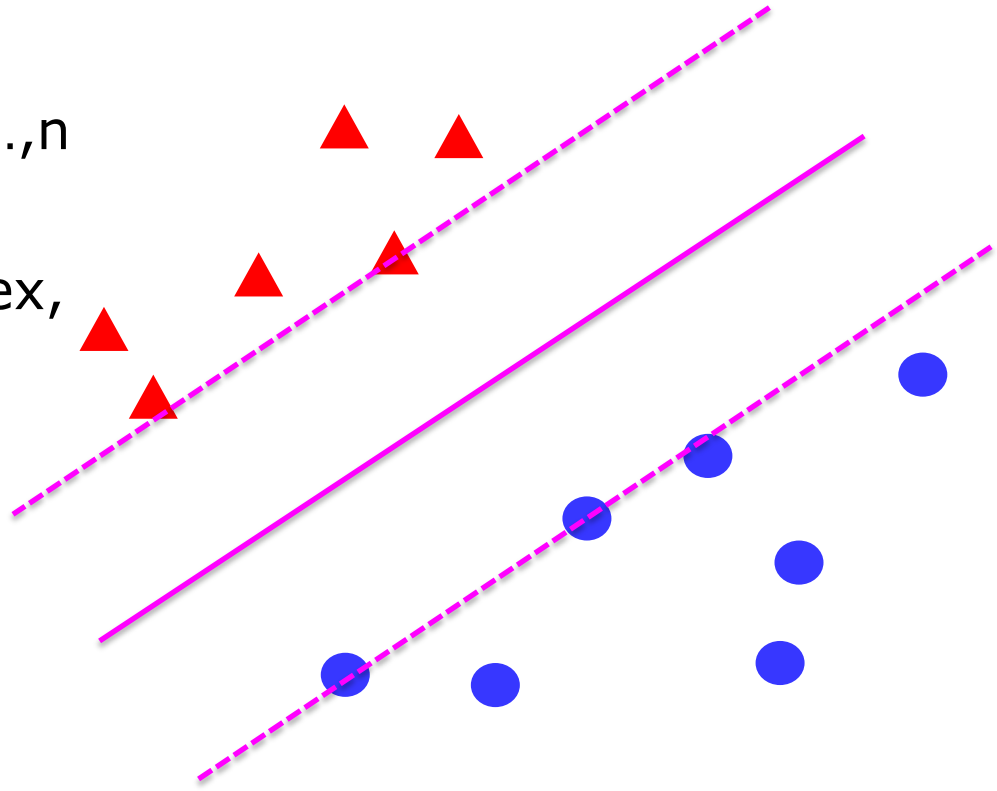
- To see $\frac{1}{2}\|w\|^2 = \frac{1}{2}\sum_{i=1}^{d} w_i^2$ convex, recall convex if Hessian matrix is PSD (for all z, $z^T Hz \geq 0$)
  - $H = \begin{pmatrix} \frac{\partial \|w\|^2}{\partial w_1 \partial w_1} & \cdots & \frac{\partial \|w\|^2}{\partial w_1 \partial w_d} \\ \vdots & \ddots & \vdots \\ \frac{\partial \|w\|^2}{\partial w_d \partial w_1} & \cdots & \frac{\partial \|w\|^2}{\partial w_d \partial w_d} \end{pmatrix}$



Rewriting Problem

- Support Vector Machines
  - $\min_{w,b} \frac{1}{2}\|w\|^2$
  - subject to:
    - $y_i(w^T x_i + b) \geq 1$ i=1,…,n

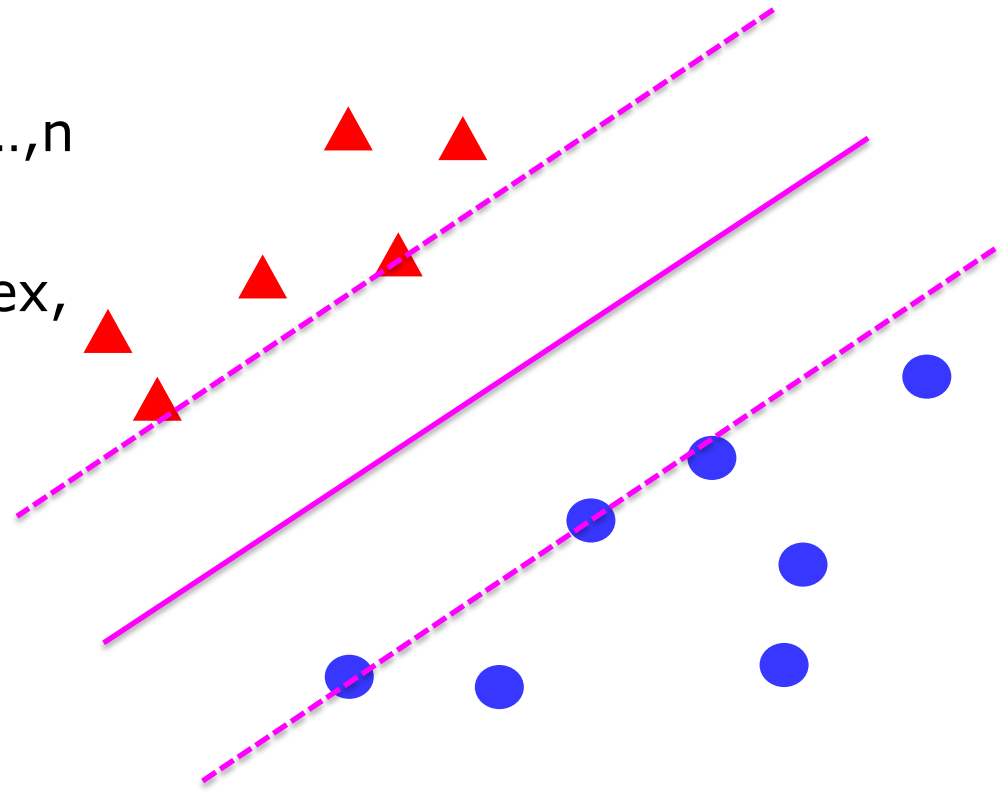
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Rewriting Problem

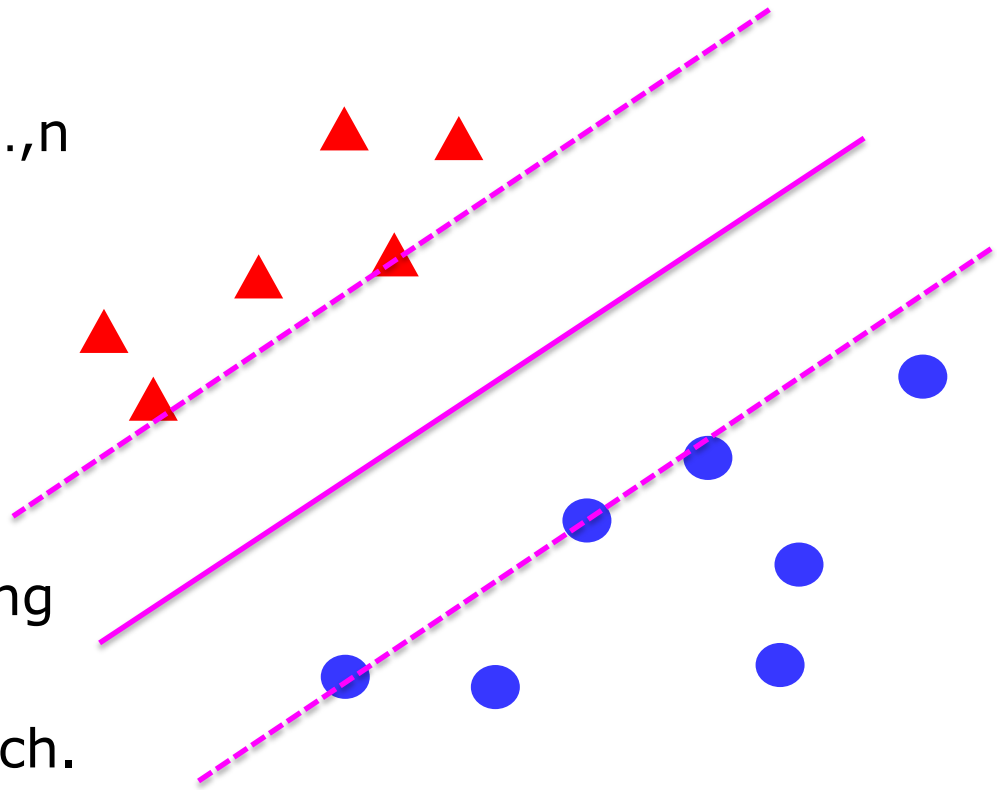
- Support Vector Machines
  - $\min_{w,b} \frac{1}{2}\|w\|^2$
  - subject to:
    - $y_i(w^T x_i + b) \geq 1$ i=1,…,n

- To see $\frac{1}{2}\|w\|^2 = \frac{1}{2}\sum_{i=1}^{d} w_i^2$ convex, recall convex if Hessian matrix is PSD (for all z, $z^T Hz \geq 0$)
  - $H = \begin{pmatrix} 1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 1 \end{pmatrix} = I$
  - $z^T Iz = z^T z = \sum_{i=1}^{d} z_i^2 \geq 0$



Rewriting Problem

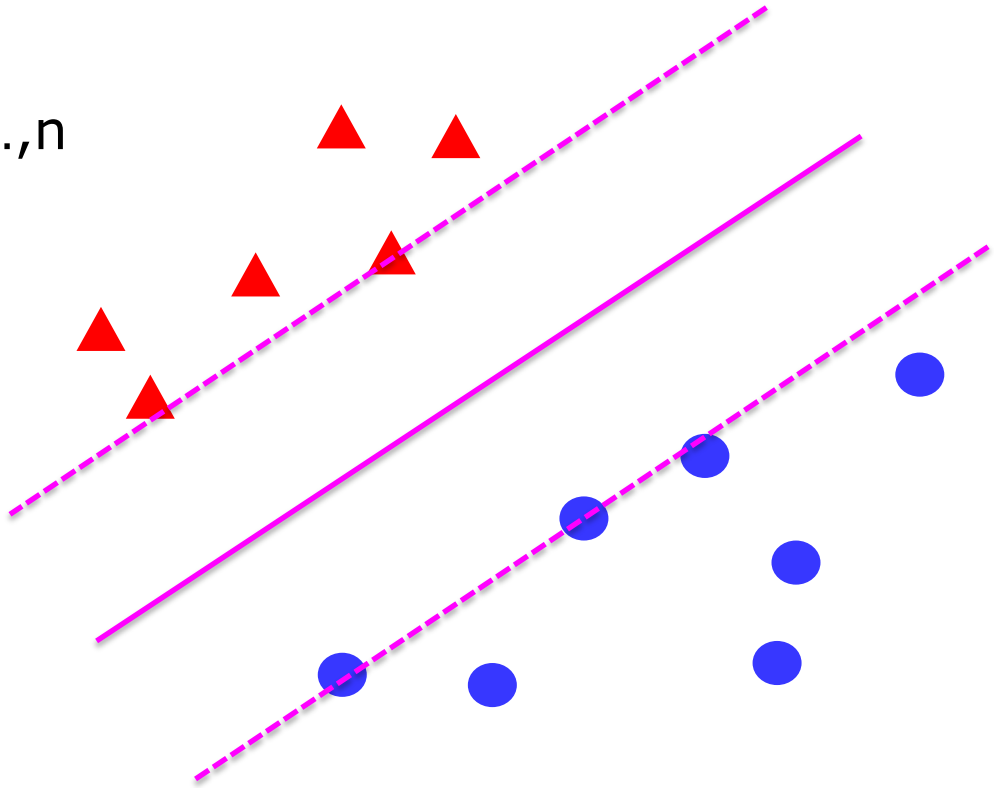
- Support Vector Machines
  - $\min_{w,b} \frac{1}{2}\|w\|^2$
  - subject to:
    - $y_i(w^T x_i + b) \geq 1$ i=1,…,n



- Minimize quadratic objective subject to linear constraints.
- Can be solved by freely available quadratic programming software.
- We will see more direct approach.
- Important for applications and more insights.

Rewriting Problem

- Support Vector Machines
  - $\min_{w,b} \frac{1}{2}\|w\|^2$
  - subject to:
    - $y_i(w^T x_i + b) \geq 1$ i=1,…,n



- Next Up:
  - Crash course on convex optimization.

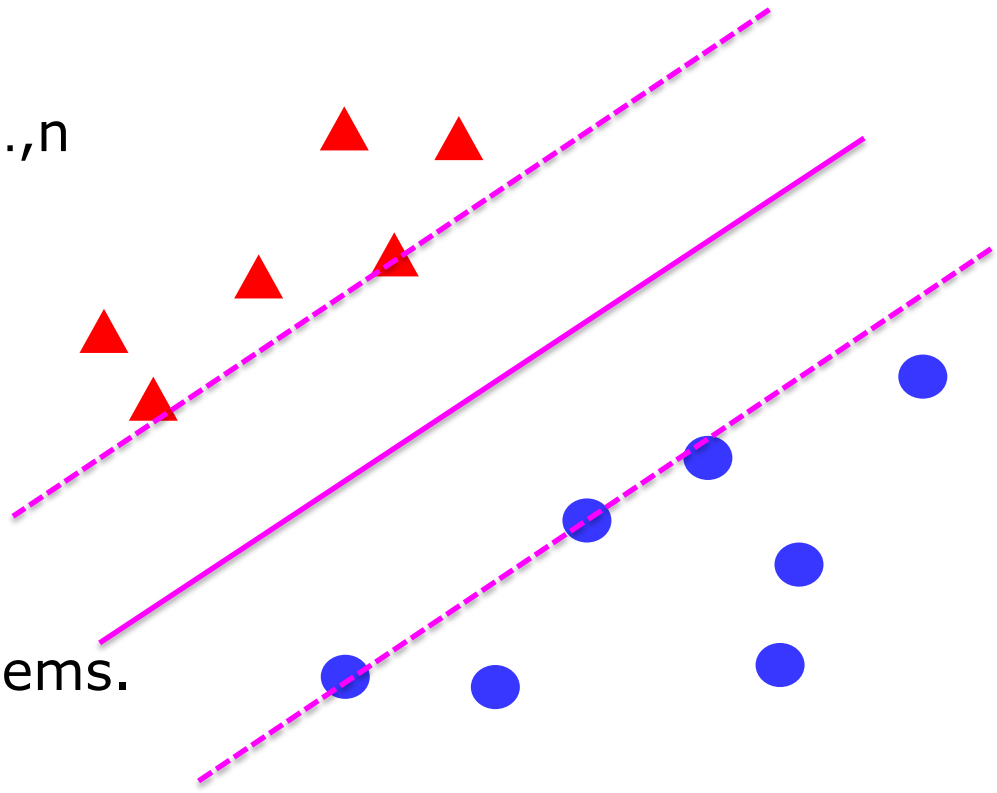
Support Vector Machines

[SVM] 1-6

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Rewriting Problem

- Support Vector Machines
  - $\min_{w,b} \frac{1}{2}\|w\|^2$
  - subject to:
    - $y_i(w^T x_i + b) \geq 1$ i=1,…,n

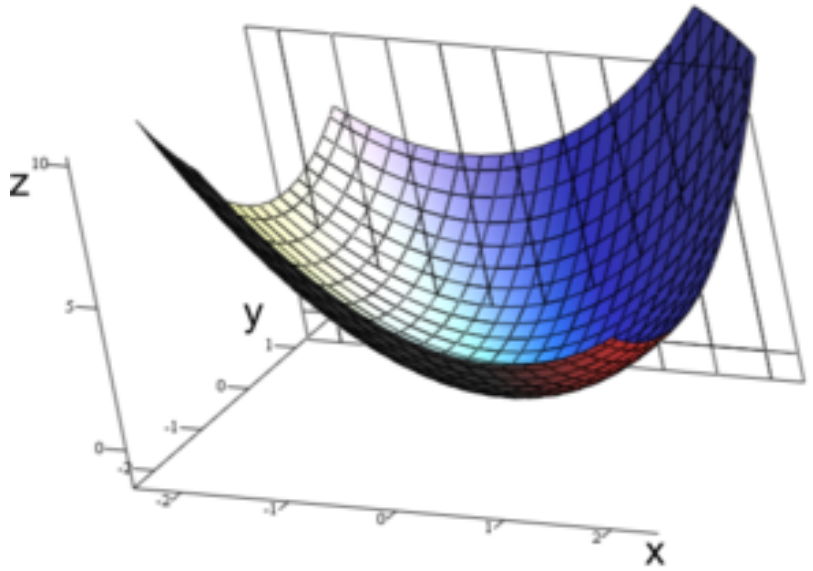


- Convex objective function.
- Linear constraints.

- We'll study general method for solving such optimization problems.

Convex Optimization

- General convex optimization problem:
  - $\min_x f(x)$
  - subject to:
    - $g_i(x) \leq 0$ i=1,…,n
  - where f and g_i's are convex functions.



Convex Optimization

- General convex optimization problem:
  - $\min_x f(x)$
  - subject to:
    - $g_i(x) \leq 0$ i=1,…,n
  - where f and g_i's are convex functions.

- Call this the primal optimization problem.

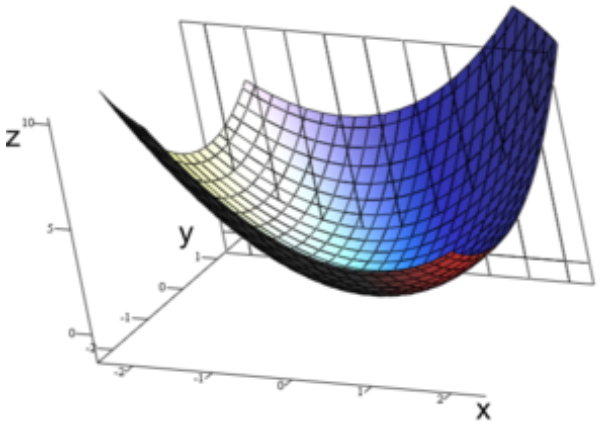
- To solve it, introduce the **generalized Lagrangian**

  - $L(x, \alpha) = f(x) + \sum_{i=1}^{n} \alpha_i g_i(x)$

- The α_i's are called **Lagrange multipliers**

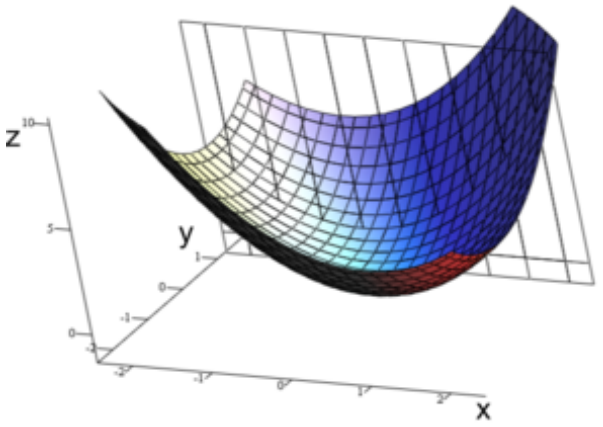


Convex Optimization



- General convex optimization problem:
  - $\min_x f(x)$
  - subject to:
    - $g_i(x) \leq 0$ i=1,…,n
- Generalized Lagrangian:
  - $L(x, \alpha) = f(x) + \sum_{i=1}^{n} \alpha_i g_i(x)$
- Consider the quantity:
  - $\theta_P(x) = \max_{\alpha:\, \alpha_i \geq 0} L(x, \alpha)$
- Assume x violates a primal constraint $g_j(x) > 0$.
  - $\theta_P(x) = \max_{\alpha:\, \alpha_i \geq 0} f(x) + \sum_{i=1}^{n} \alpha_i g_i(x)$
  - Can let $\alpha_j \to \infty$ and thus $\theta_P(x) = \infty$

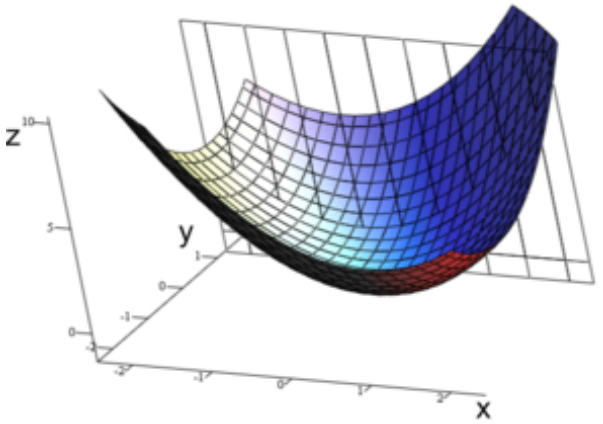
Convex Optimization



- General convex optimization problem:
  - $\min_x f(x)$
  - subject to:
    - $g_i(x) \leq 0$ i=1,…,n
- Generalized Lagrangian:
  - $L(x, \alpha) = f(x) + \sum_{i=1}^{n} \alpha_i g_i(x)$
- Consider the quantity:
  - $\theta_P(x) = \max_{\alpha:\, \alpha_i \geq 0} L(x, \alpha)$
- Assume x satisfies all primal constraints $g_j(x) \leq 0$ for all j.
  - $\theta_P(x) = \max_{\alpha:\, \alpha_i \geq 0} f(x) + \sum_{i=1}^{n} \alpha_i g_i(x)$
  - Best to set all $\alpha_j = 0$ and get $\theta_P(x) = f(x)$



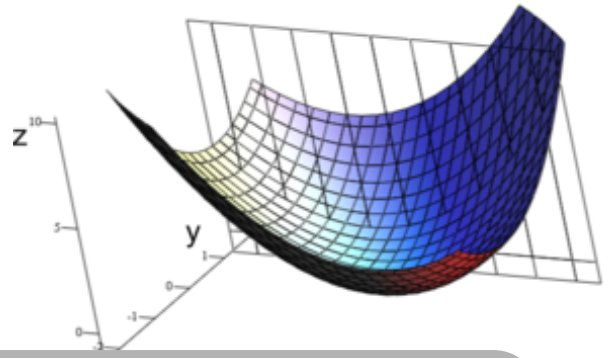
Convex Optimization



- General convex optimization problem:
  - $\min_x f(x)$
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    - $g_i(x) \leq 0$ i=1,…,n
- Generalized Lagrangian:
  - $L(x, \alpha) = f(x) + \sum_{i=1}^{n} \alpha_i g_i(x)$
- Consider the quantity:
  - $\theta_P(x) = \max_{\alpha:\, \alpha_i \geq 0} L(x, \alpha)$
- Observations:
  - $\theta_P(x) = \infty$ if x violates primal constraint.
  - $\theta_P(x) = f(x)$ if x satisfies all primal constraints.
  - $\min_x \theta_P(x) = \min_x \max_{\alpha:\, \alpha_i \geq 0} L(x, \alpha)$ same solution as primal.



Convex Optimization



- General convex optimization problem:
  - $\min_x f(x)$
  - subject to:
    - $g_i(x) \leq 0$ i=1,…,n
- Generalized Lagrangian:
  - $L(x,\alpha) = f(x) + \sum_{i=1}^{n} \alpha_i g_i(x)$
  - $\theta_P(x) = \max_{\alpha:\, \alpha_i \geq 0} L(x,\alpha)$
  - $\min_x \theta_P(x) = \min_x \max_{\alpha:\, \alpha_i \geq 0} L(x,\alpha)$ same optimization problem.

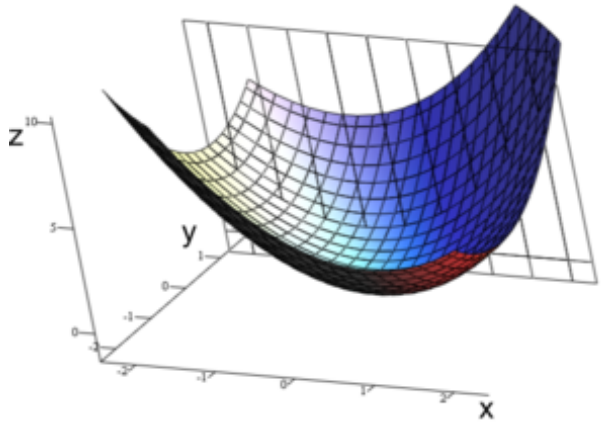
Downside:

Introduced more variables and a max.

Achieved:

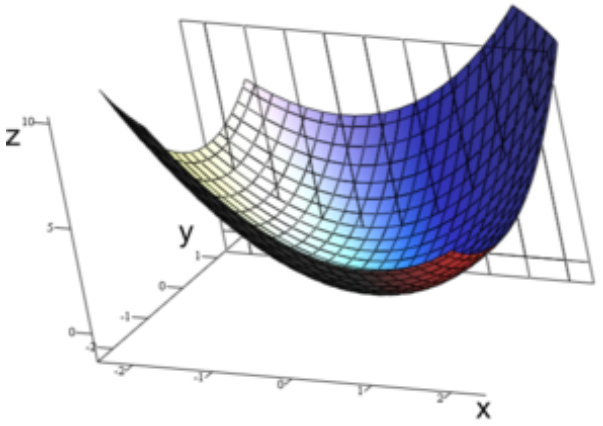
Unconstrained optimization problem!

Convex Optimization



- General convex optimization problem:
  - $\min_x f(x)$
  - subject to:
    - $g_i(x) \leq 0$ i=1,…,n
- Generalized Lagrangian:
  - $L(x,\alpha) = f(x) + \sum_{i=1}^{n} \alpha_i g_i(x)$
  - $\theta_P(x) = \max_{\alpha:\, \alpha_i \geq 0} L(x,\alpha)$
  - $\min_x \theta_P(x) = \min_x \max_{\alpha:\, \alpha_i \geq 0} L(x,\alpha)$ same optimization problem.
- Primal value:
  - Let $p^* = \min_x \theta_P(x)$ denote the **primal value**.

Convex Optimization



- Generalized Lagrangian:
  - $L(x, \alpha) = f(x) + \sum_{i=1}^{n} \alpha_i g_i(x)$
  - $\theta_P(x) = \max_{\alpha:\, \alpha_i \geq 0} L(x, \alpha)$
  - $p^* = \min_x \theta_P(x) = \min_x \max_{\alpha:\, \alpha_i \geq 0} L(x, \alpha)$
- Dual Problem:
  - $\theta_D(\alpha) = \min_x L(x, \alpha)$
  - $d^* = \max_{\alpha:\, \alpha_i \geq 0} \theta_D(\alpha) = \max_{\alpha:\, \alpha_i \geq 0} \min_x L(x, \alpha)$
  - Called **dual value**.

Flipped order

- Observation:
  - $d^* \leq p^*$

Max-Min vs. Min-Max

- Function of two variables.
  - $L(x, \alpha)$
- Claim:
  - $\max_{\alpha \in A} \min_{x \in X} L(x, \alpha) \leq \min_{x \in X} \max_{\alpha \in A} L(x, \alpha)$
- Proof by Picture:

Smallest value in row α

A \ X				
0	2	4	8	1
4	6	3	5	9
1	3	5	8	7
9	2	1	3	5
4	7	2	6	8



Max-Min vs. Min-Max

- Function of two variables.
  - $L(x, \alpha)$
- Claim:
  - $\max_{\alpha \in A} \min_{x \in X} L(x, \alpha) \leq \min_{x \in X} \max_{\alpha \in A} L(x, \alpha)$
- Proof by Picture:

Largest minimum value in a row

X

A

0	2	4	8	1
4	6	3	5	9
1	3	5	8	7
9	2	1	3	5
4	7	2	6	8



Max-Min vs. Min-Max

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- Proof by Picture:

Largest value in column x

X

A

0	2	4	8	1
4	6	3	5	9
1	3	5	8	7
9	2	1	3	5
4	7	2	6	8



Max-Min vs. Min-Max

- Function of two variables.
  - $L(x, \alpha)$
- Claim:
  - $\max_{\alpha \in A} \min_{x \in X} L(x, \alpha) \leq \min_{x \in X} \max_{\alpha \in A} L(x, \alpha)$
- Proof by Picture:

Smallest maximum value in a column

X

A

0	2	4	8	1
4	6	3	5	9
1	3	5	8	7
9	2	1	3	5
4	7	2	6	8

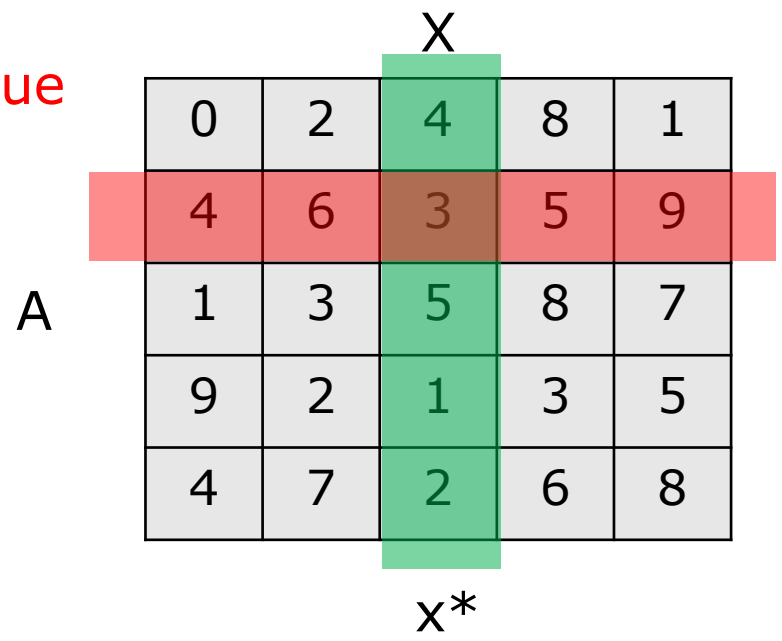


Max-Min vs. Min-Max

- Function of two variables.
  - $L(x, \alpha)$
- Claim:
  - $\max_{\alpha \in A} \min_{x \in X} L(x, \alpha) \leq \min_{x \in X} \max_{\alpha \in A} L(x, \alpha)$

Smallest maximum value in a column

Largest minimum value in a row

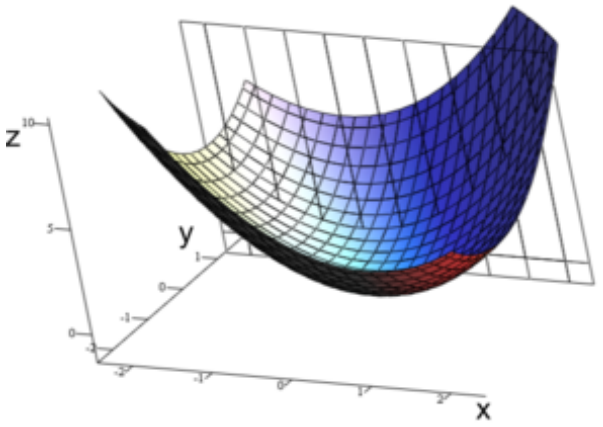


Every row *a* intersects column x* with smallest max value.

For every row *a*, the entry in col. x* is $\leq \max_{\alpha} L(x^*, \alpha)$

So for every row *a*, min value $\leq \max_{\alpha} L(x^*, \alpha)$

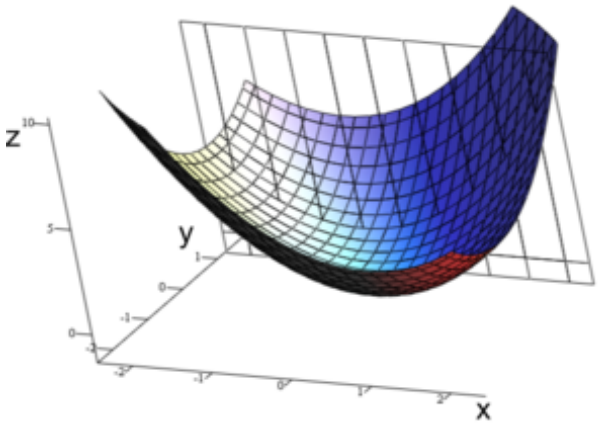
Convex Optimization



- Generalized Lagrangian:
  - $L(x, \alpha) = f(x) + \sum_{i=1}^{n} \alpha_i g_i(x)$
  - $\theta_P(x) = \max_{\alpha:\, \alpha_i \geq 0} L(x, \alpha)$
  - $p^* = \min_x \theta_P(x) = \min_x \max_{\alpha:\, \alpha_i \geq 0} L(x, \alpha)$
- Dual Problem:
  - $\theta_D(\alpha) = \min_x L(x, \alpha)$
  - $d^* = \max_{\alpha:\, \alpha_i \geq 0} \theta_D(\alpha) = \max_{\alpha:\, \alpha_i \geq 0} \min_x L(x, \alpha)$
  - $d^* \leq p^*$
- Idea:
  - If $d^* = p^*$ then we can solve dual problem instead of primal problem.



Convex Optimization



- Generalized Lagrangian:
  - $L(x, \alpha) = f(x) + \sum_{i=1}^{n} \alpha_i g_i(x)$
  - $\theta_P(x) = \max_{\alpha:\, \alpha_i \geq 0} L(x, \alpha)$
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- Dual Problem:
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  - $d^* = \max_{\alpha:\, \alpha_i \geq 0} \theta_D(\alpha) = \max_{\alpha:\, \alpha_i \geq 0} \min_x L(x, \alpha)$
- Result:
  - If f and g_i's are convex functions and the constraints g_i are strictly feasible (there is an x with $g_i(x) < 0$ for all i), then
    - $d^* = p^*$



Convex Optimization

- Result:
  - If f and g_i's are convex functions and the constraints g_i are strictly feasible (there is an x with $g_i(x) < 0$ for all i), then
    - $d^* = p^*$

- Wait, there is more!
- **Karush-Kuhn-Tucker** conditions:
  - There is an optimal solution x*,α* satisfying:
    - $\frac{\partial L(x^*,\alpha^*)}{\partial x_i} = 0$ for i=1,…,d
    - $\alpha_i^* g_i(x^*) = 0$ for i=1,…,n
    - $g_i(x^*) \leq 0$ for i=1,…,n
    - $\alpha_i^* \geq 0$ for i=1,…,n



Convex Optimization

- Result:
  - If f and g_i's are convex functions and the constraints g_i are strictly feasible (there is an x with $g_i(x) < 0$ for all i), then
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- Wait, there is more!
- **Karush-Kuhn-Tucker** conditions:
  - There is an optimal solution x*,α* satisfying:
    - $\frac{\partial L(x^*,\alpha^*)}{\partial x_i} = 0$ for i=1,…,d
    - $\alpha_i^* g_i(x^*) = 0$ for i=1,…,n
    - $g_i(x^*) \leq 0$ for i=1,…,n
    - $\alpha_i^* \geq 0$ for i=1,…,n

Important:

If $\alpha_i^* > 0$ then must have $g_i(x^*) = 0$. Means: $g_i(x) \leq 0$ an active constraint.



Convex Optimization

- Result:
  - If f and g_i's are convex functions and the constraints g_i are strictly feasible (there is an x with $g_i(x) < 0$ for all i), then
    - $d^* = p^*$
    - There is an optimal solution x*,α* satisfying:
      - $\frac{\partial L(x^*,\alpha^*)}{\partial x_i} = 0$ for i=1,…,d
      - $\alpha_i^* g_i(x^*) = 0$ for i=1,…,n
      - $g_i(x^*) \leq 0$ for i=1,…,n
      - $\alpha_i^* \geq 0$ for i=1,…,n
- Implications:
  - Can solve dual problem, and, assume solution satisfies all of the above.

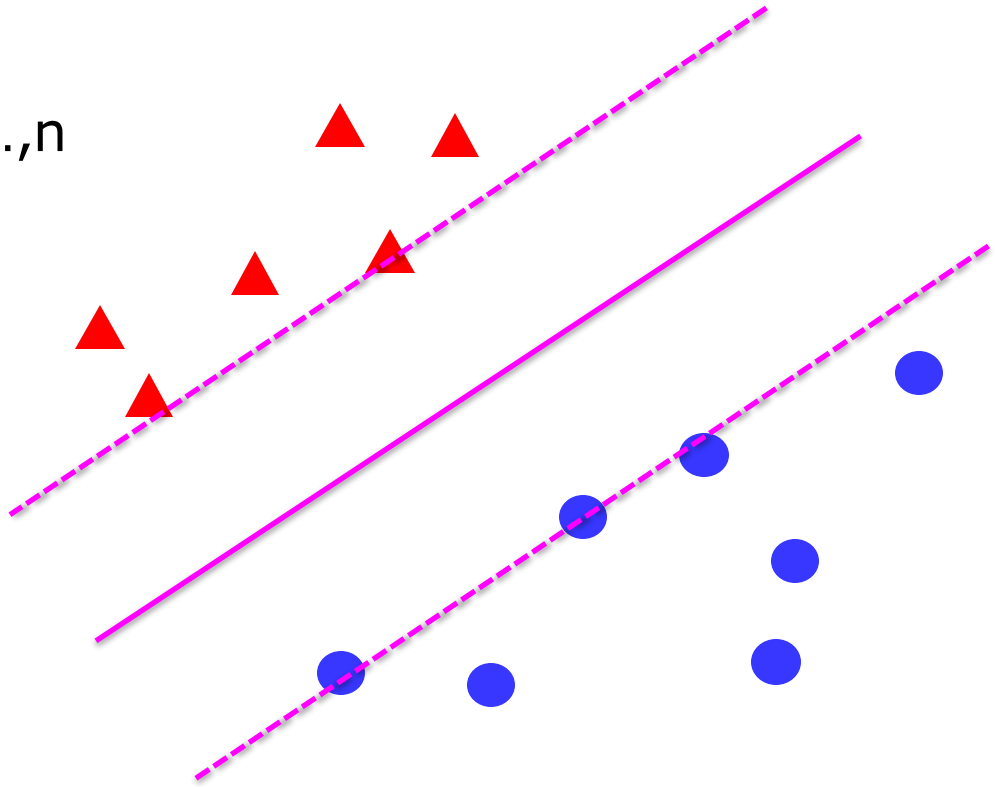


Support Vector Machines

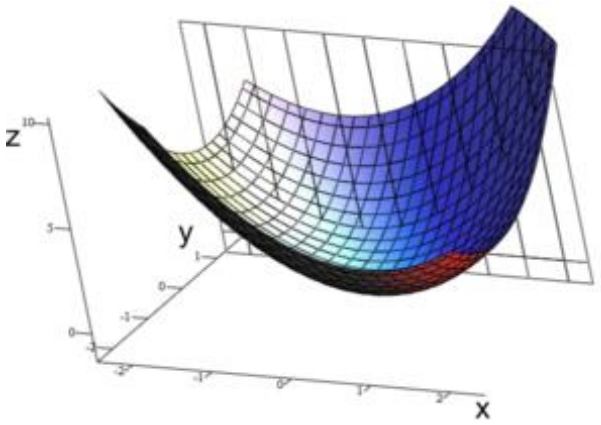
Rewriting Problem

- Support Vector Machines
  - $\min_{w,b} \frac{1}{2}\|w\|^2$
  - subject to:
    - $y_i(w^T x_i + b) \geq 1$ i=1,…,n

- Convex objective function.
- Linear constraints.



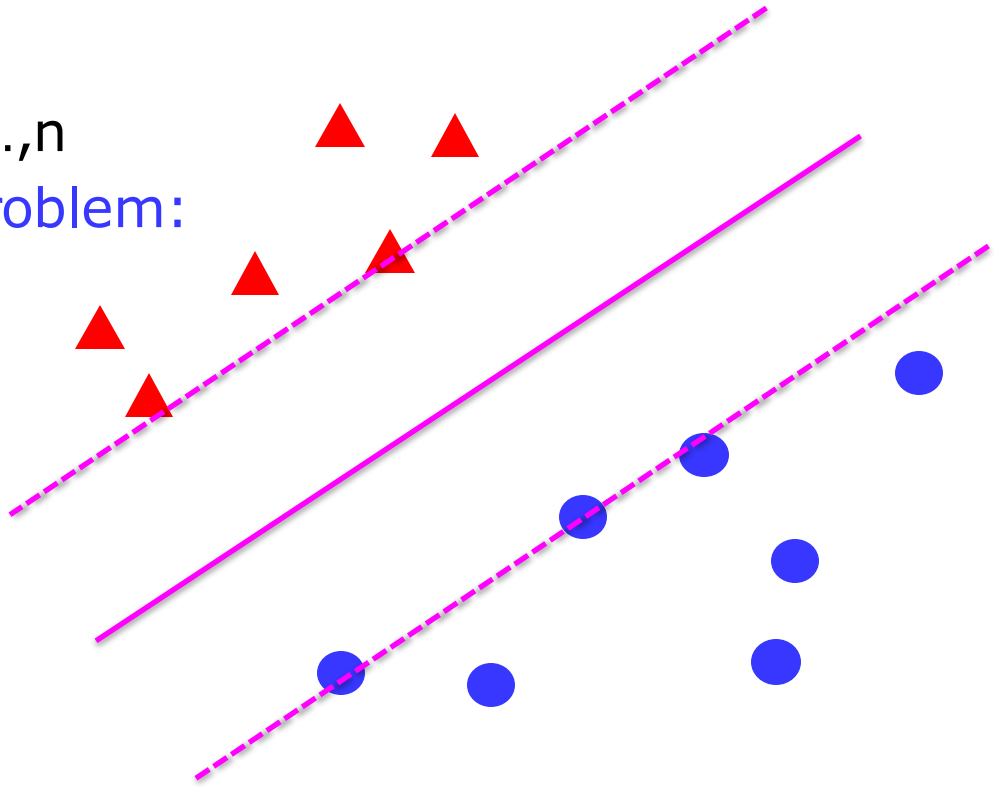
Convex Optimization



- General convex optimization problem:
  - $\min_x f(x)$
  - subject to:
    - $g_i(x) \leq 0$ i=1,…,n
- Generalized Lagrangian:
  - $L(x,\alpha) = f(x) + \sum_{i=1}^{n} \alpha_i g_i(x)$
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  - $d^* = \max_{\alpha:\, \alpha_i \geq 0} \theta_D(\alpha) = \max_{\alpha:\, \alpha_i \geq 0} \min_x L(x,\alpha)$

Rewriting Problem

- Support Vector Machines
  - $\min_{w,b} \frac{1}{2}\|w\|^2$
  - subject to:
    - $y_i(w^T x_i + b) \geq 1$ i=1,…,n
- General convex optimization problem:
  - $\min_x f(x)$
  - subject to:
    - $g_i(x) \leq 0$ i=1,…,n
- SVM in "standard" form:
  - $f(w,b) = \frac{1}{2}\|w\|^2$
    - $g_i(w,b) \leq 0$, with
    - $g_i(w,b) = 1 - y_i(w^T x_i + b)$



Convex Optimization

- (KKT) Karush-Kuhn-Tucker conditions:
  - If f and g_i's are convex functions and the constraints g_i are strictly feasible (there is an x with $g_i(x) < 0$ for all i), then
    - $d^* = p^*$
    - There is an optimal solution x*,α* satisfying:
      - $\frac{\partial L(x^*,\alpha^*)}{\partial x_i} = 0$ for i=1,…,d
      - $\alpha_i^* g_i(x^*) = 0$ for i=1,…,n
      - $g_i(x^*) \leq 0$ for i=1,…,n
      - $\alpha_i^* \geq 0$ for i=1,…,n
- Implications:
  - Can solve dual problem, and, assume solution satisfies all of the above.

Rewriting Problem

- SVM in "standard" form:
  - $\min_{w,b} f(w,b)$, with $f(w,b) = ½\|w\|^2$
  - subject to:
    - $g_i(w,b) \leq 0$, with $g_i(w,b) = 1 - y_i(w^T x_i + b)$

- Already showed $f(w,b)$ is convex. Need to show g_i convex.
- Hessian H_i

$$H_i = \begin{pmatrix} \frac{\partial^2(1-y_i(w^T x_i+b))}{\partial w_1 \partial w_1} & \cdots & \frac{\partial^2(1-y_i(w^T x_i+b))}{\partial w_1 \partial b} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2(1-y_i(w^T x_i+b))}{\partial b \partial w_1} & \cdots & \frac{\partial^2(1-y_i(w^T x_i+b))}{\partial b \partial b} \end{pmatrix}$$

Rewriting Problem

- SVM in "standard" form:
  - $\min_{w,b} f(w,b)$, with $f(w,b) = ½\|w\|^2$
  - subject to:
    - $g_i(w,b) \leq 0$, with $g_i(w,b) = 1 - y_i(w^T x_i + b)$

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- Already showed $f(w,b)$ is convex. Need to show g_i convex.
- Hessian H_i

$$H_i = \begin{pmatrix} 0 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 0 \end{pmatrix}$$

- Clearly $z^T H_i z = 0 \geq 0$ for all z.
- So convex.

Rewriting Problem

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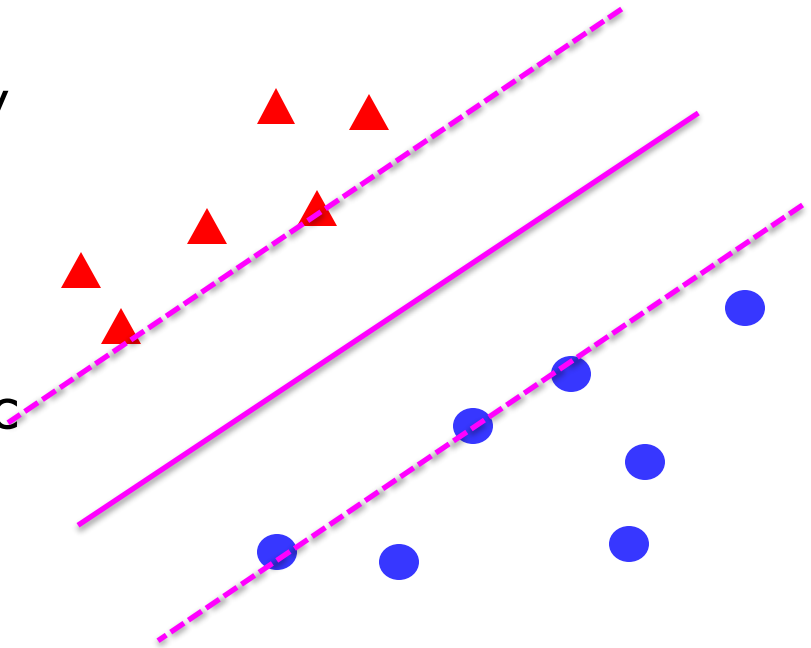
Convex Optimization

- (KKT) Karush-Kuhn-Tucker conditions:
  - If f and g_i's are convex functions and the constraints g_i are strictly feasible (there is an x with $g_i(x) < 0$ for all i), then
    - $d^* = p^*$
    - There is an optimal solution x*,α* satisfying:
      - $\frac{\partial L(x^*,\alpha^*)}{\partial x_i} = 0$ for i=1,…,d
      - $\alpha_i^* g_i(x^*) = 0$ for i=1,…,n
      - $g_i(x^*) \leq 0$ for i=1,…,n
      - $\alpha_i^* \geq 0$ for i=1,…,n
- Implications:
  - Can solve dual problem, and, assume solution satisfies all of the above.

Rewriting Problem

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- If data linearly separable, pick any separating hyperplane $w^T x+b$.
- Correctly classifies all, so $y_i(w^T x_i + b) > 0$ for all i.
- Scale w and b by arbitrarily large c to make $g_i(w,b) < 0$.





Convex Optimization

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    - $g_i(w,b) \leq 0$, with $g_i(w,b) = 1 - y_i(w^T x_i + b)$

- Instead solve dual:
  - $\max_{\alpha:\, \alpha_i \geq 0} \min_{w,b} f(w,b) + \sum_{i=1}^{n} \alpha_i g_i(w,b)$
  - $\max_{\alpha:\, \alpha_i \geq 0} \min_{w,b} ½\|w\|^2 - \sum_{i=1}^{n} \alpha_i (y_i(w^T x_i + b) - 1)$

Convex Optimization

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- Instead solve dual:
  - $\max_{\alpha:\, \alpha_i \geq 0} \min_{w,b} ½\|w\|^2 - \sum_{i=1}^n \alpha_i (y_i(w^T x_i + b) - 1)$

- Setting $\frac{\partial ½\|w\|^2 - \sum_{i=1}^n \alpha_i (y_i(w^T x_i + b) - 1)}{\partial w_j} = 0$:
  - $\frac{\partial ½\|w\|^2 - \sum_{i=1}^n \alpha_i (y_i(w^T x_i + b) - 1)}{\partial w_j} = w_j - \sum_{i=1}^n \alpha_i y_i x_{i,j} = 0$
  - $w = \sum_{i=1}^n \alpha_i y_i x_i$

Rewriting Problem

- SVM in "standard" form:
  - $\min_{w,b} f(w,b)$, with $f(w,b) = ½\|w\|^2$
  - subject to:
    - $g_i(w,b) \leq 0$, with $g_i(w,b) = 1 - y_i(w^T x_i + b)$

- Instead solve dual:
  - $\max_{\alpha:\, \alpha_i \geq 0} \min_{w,b} ½\|\sum_{i=1}^n \alpha_i y_i x_i\|^2 - \sum_{i=1}^n \alpha_i (y_i \left((\sum_{j=1}^n \alpha_j y_j x_j)^T x_i + b \right) - 1)$

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  - $\max_{\alpha:\, \alpha_i \geq 0} \min_{w,b} ½\|\sum_{i=1}^n \alpha_i y_i x_i\|^2 - \sum_{i=1}^n \alpha_i (y_i \left((\sum_{j=1}^n \alpha_j y_j x_j)^T x_i + b \right) - 1)$
  - $\max_{\alpha:\, \alpha_i \geq 0} \min_{w,b} ½ \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j x_i^T x_j - \sum_{i=1}^n \alpha_i (y_i \left((\sum_{j=1}^n \alpha_j y_j x_j)^T x_i + b \right) - 1)$
  - $\max_{\alpha:\, \alpha_i \geq 0} \min_{w,b} \sum_{i=1}^n \alpha_i - ½ \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j x_i^T x_j - b \sum_{i=1}^n \alpha_i y_i$

Rewriting Problem

- SVM in "standard" form:
  - $\min_{w,b} f(w,b)$, with $f(w,b) = ½\|w\|^2$
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- Setting $\frac{\partial ½\|w\|^2 - \sum_{i=1}^n \alpha_i (y_i (w^T x_i + b) - 1)}{\partial b} = 0$:
  - $\frac{\partial ½\|w\|^2 - \sum_{i=1}^n \alpha_i (y_i (w^T x_i + b) - 1)}{\partial b} = -\sum_{i=1}^n \alpha_i y_i = 0$
  - $\sum_{i=1}^n \alpha_i y_i = 0$

Rewriting Problem

- SVM in "standard" form:
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  - subject to:
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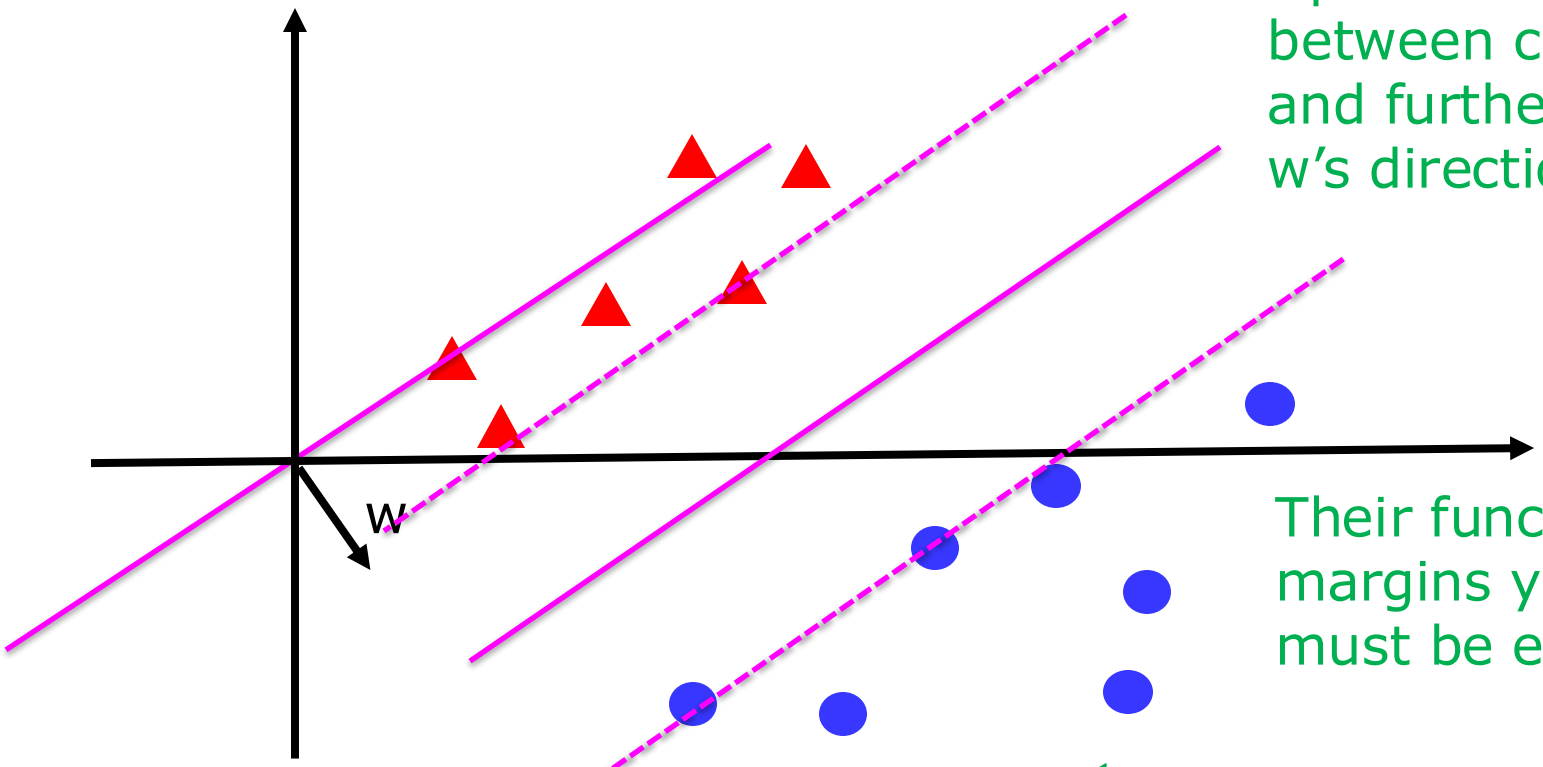
- No w and b in expression, so same as:
  - $\max_{\alpha:\, \alpha_i \geq 0} \sum_{i=1}^n \alpha_i - ½ \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j x_i^T x_j$

Rewriting Problem

- SVM optimization problem:
  - $\max_{\alpha:\, \alpha_i \geq 0} \sum_{i=1}^{n} \alpha_i - \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} \alpha_i \alpha_j y_i y_j x_i^T x_j$
  - subject to:
    - $\sum_{i=1}^{n} \alpha_i y_i = 0$
- Once solved, solution hyperplane satisfies:
  - $w = \sum_{i=1}^{n} \alpha_i y_i x_i$
- How do we recover optimal b?

Rewriting Problem

- Recovering intercept b of optimal hyperplane w.



Optimal is mid-way between closest blue and furthest red in w's direction.

Their functional margins $y_i(w^T x_i + b)$ must be equal.

$$\min_{i:y_i=1} w^T x_i + b = -(\max_{i:y_i=-1} w^T x_i + b) \qquad b = -\frac{1}{2}(\min_{i:y_i=1} w^T x_i + \max_{i:y_i=-1} w^T x_i)$$

Rewriting Problem

- SVM optimization problem:
  - $\max_{\alpha:\, \alpha_i \geq 0} \sum_{i=1}^{n} \alpha_i - ½ \sum_{i=1}^{n} \sum_{j=1}^{n} \alpha_i \alpha_j y_i y_j x_i^T x_j$
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Convex Optimization

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Rewriting Problem

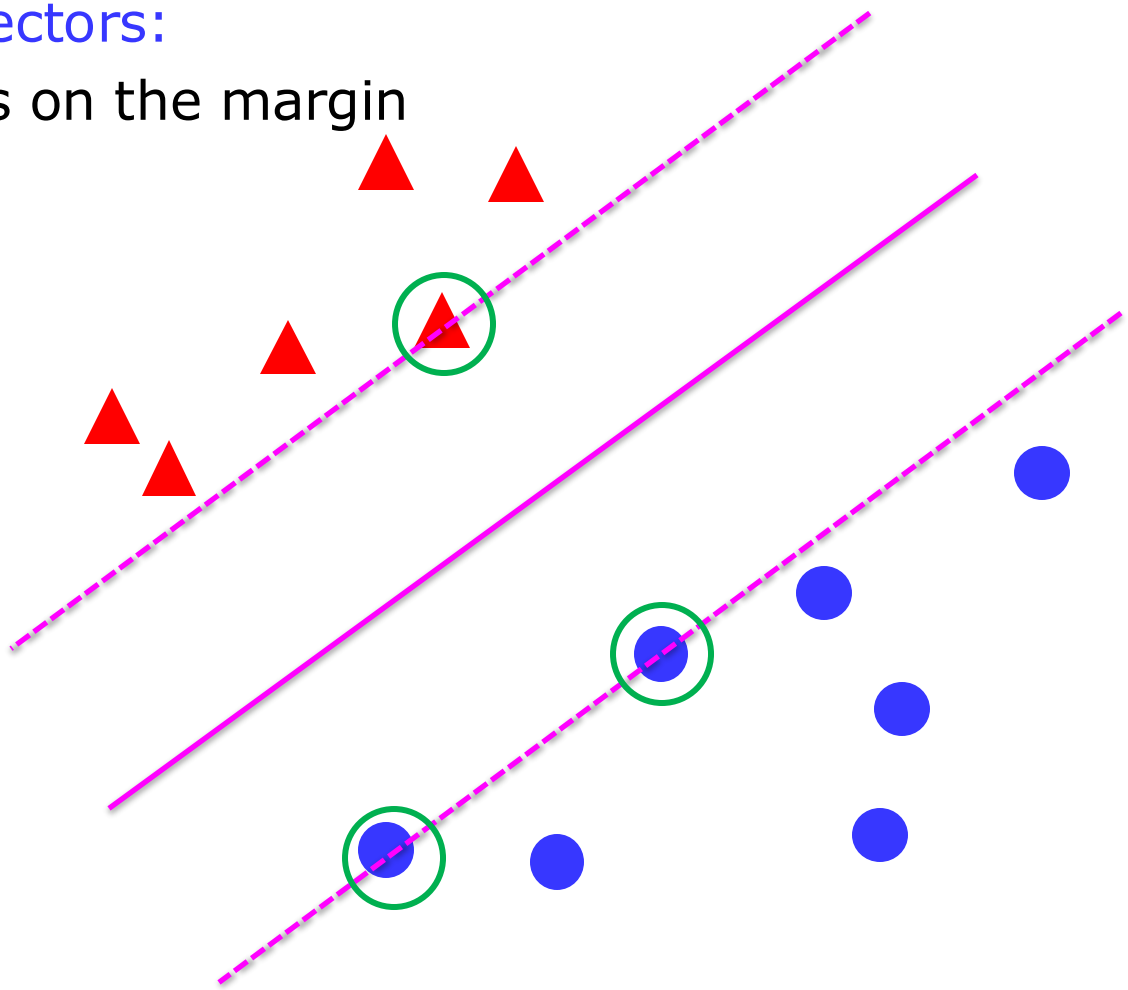
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  - $w = \sum_{i=1}^{n} \alpha_i y_i x_i$
  - $b = -\frac{1}{2}\left(\min_{i:y_i=1} w^T x_i + \max_{i:y_i=-1} w^T x_i\right)$

- The value α_i is only non-zero when $y_i(w^T x_i + b) = 1$.
  - Means (x_i, y_i) is on the margin!

Rewriting Problem

- Support Vectors:
  - Vectors on the margin



Rewriting Problem

- SVM optimization problem:
  - $\max_{\alpha:\, \alpha_i \geq 0} \sum_{i=1}^{n} \alpha_i - \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} \alpha_i \alpha_j y_i y_j x_i^T x_j$
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  - $b = -\frac{1}{2}(\min_{i:y_i=1} w^T x_i + \max_{i:y_i=-1} w^T x_i)$

- w is linear combination of only the support vectors.
- Very important in next lecture!
- Motivates the name ”Support Vector Machines”.